



National and Kapodistrian University of Athens (NKUA)
MSc in Data Science and Information Technologies (DSIT)
Course: Knowledge Technologies (2025–2026)

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Exercise 1

1)

To answer the question “Find all Greek wines known by DBpedia and the region of Greece where they are produced” I queried the public DBpedia SPARQL endpoint (<http://dbpedia.org/sparql>).

In this context, I consider as Greek wines all DBpedia resources that are linked to a wine region whose country is Greece. Formally, these are resources ?wine that are related via the property **dbo:wineRegion** to some region ?region such that ?region **dbo:country dbr:Greece**. For each wine and region I also retrieve the English labels. In addition, I include a second UNION block that captures Greek wines whose regional information is not expressed via **dbo:wineRegion**, but instead through alternative properties such as **dbo:region**, **dbp:region**, **dbo:origin**, or **dbp:origin**. This block selects resources typed as **dbo:Wine**, explicitly marked with **dbo:country dbr:Greece**, and associated with a region whose country is also Greece. It is added to ensure completeness, since DBpedia often uses heterogeneous properties for geographic origin; therefore, it guarantees that all Greek wines with valid Greek production regions are retrieved, even when the ontology does not employ **dbo:wineRegion**.

The SPARQL query I used is:

```
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX dbp: <http://dbpedia.org/property/>
PREFIX dbr: <http://dbpedia.org/resource/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT DISTINCT ?wine ?wineLabel ?region ?regionLabel
WHERE {
  {
    # CASE 1: wineRegion → region in Greece
    ?wine dbo:wineRegion ?region .
    ?region dbo:country dbr:Greece .
  }
  UNION
  {
    # CASE 2: typed as Wine, Greek country, region/ origin multiple properties
    ?wine rdf:type dbo:Wine .
    ?wine dbo:country dbr:Greece .
  }
}
```

```

?wine (dbo:region | dbp:region | dbo:origin | dbp:origin) ?region .
?region dbo:country dbr:Greece .
}
# labels
OPTIONAL {
?wine rdfs:label ?wineLabel .
FILTER (lang(?wineLabel) = "en")
}
OPTIONAL {
?region rdfs:label ?regionLabel .
FILTER (lang(?regionLabel) = "en")
}
}
ORDER BY ?wineLabel

```

Executing this query on the DBpedia endpoint returns the following table:

SPARQL HTML5 table			
wine	wineLabel	region	regionLabel
http://dbpedia.org/resource/Chios	"Chios" ^{en}	http://dbpedia.org/resource/Thrace	"Thrace" ^{en}
http://dbpedia.org/resource/Chios ^{en}	"Chios" ^{en}	http://dbpedia.org/resource/Thrace	"Thrace" ^{en}
http://dbpedia.org/resource/Chios	"Chios" ^{en}	http://dbpedia.org/resource/Thrace	"Thrace" ^{en}
http://dbpedia.org/resource/Chios	"Chios" ^{en}	http://dbpedia.org/resource/Thrace ^{en}	"Thrace, Thracian, Thracian" ^{en}
http://dbpedia.org/resource/Chios	"Chios" ^{en}	http://dbpedia.org/resource/Thrace	"Thrace" ^{en}
http://dbpedia.org/resource/Chios	"Chios" ^{en}	http://dbpedia.org/resource/Thrace ^{en}	"Thrace, Thracian" ^{en}

Figure 1: Results of the SPARQL query returning DBpedia wine-related instances, including their English labels, regions of origin etc.

2)

To find all the prime ministers of Greece known to DBpedia, one can treat as “prime ministers of Greece” all DBpedia resources that are persons and are linked from the page `dbr:List_of_prime_ministers_of_Greece` via `dbo:wikiPageWikiLink`. For each such person `?pm`, the query above retrieves:

- their English label `?name` (via `rdfs:label` with `lang = "en"`),
- all political parties they have been members of, by following `dbp:party` to a party resource and returning its English label(s), aggregated with `GROUP_CONCAT` into the field `?parties`, and
- all universities they have graduated from, by following `dbp:almaMater` to a university resource and returning its English label(s), aggregated with `GROUP_CONCAT` into the field `?universities`.

Thus, the result of this query is a table where each row corresponds to a prime minister of Greece in DBpedia and contains their name, the set of their parties, and the set of universities from which they graduated.

The query is:

```

PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX dbp: <http://dbpedia.org/property/>
PREFIX dbr: <http://dbpedia.org/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?pm ?name
  (GROUP_CONCAT(DISTINCT ?partyLabel; separator=", ") AS ?parties)
  (GROUP_CONCAT(DISTINCT ?univLabel; separator=", ") AS ?universities)
WHERE {
# Prime ministers of Greece: persons linked from the list page
dbr:List_of_prime_ministers_of_Greece dbo:wikiPageWikiLink ?pm .
?pm a dbo:Person .

```

```

# English name
OPTIONAL {
  ?pm rdfs:label ?name .
  FILTER (lang(?name) = "en")
}

# Political parties
OPTIONAL {
  ?pm dbp:party ?party .
  OPTIONAL {
    ?party rdfs:label ?partyLabel .
    FILTER (lang(?partyLabel) = "en")
  }
}

# Universities (alma mater)
OPTIONAL {
  ?pm dbp:almaMater ?univ .
  OPTIONAL {
    ?univ rdfs:label ?univLabel .
    FILTER (lang(?univLabel) = "en")
  }
}
}
GROUP BY ?pm ?name
ORDER BY ?name

```

Results of the query

pm,"name","parties","universities"

http://dbpedia.org/resource/Adamantios_Androutsopoulos,"Adamantios Androutsopoulos",",",""

http://dbpedia.org/resource/Alexandros_Diomidis,"Alexandros Diomidis","Liberal Party (Greece)",""

http://dbpedia.org/resource/Alexandros_Koryzis,"Alexandros Koryzis","Independent politician",""

http://dbpedia.org/resource/Alexandros_Koumoundouros,"Alexandros Koumoundouros","National Party (Greece)",""

http://dbpedia.org/resource/Alexandros_Mavrokordatos,"Alexandros Mavrokordatos","English Party","University of Athens"

http://dbpedia.org/resource/Alexandros_Mazarakis-Ainian,"Alexandros Mazarakis-Ainian",",",""

http://dbpedia.org/resource/Alexandros_Othonaios,"Alexandros Othonaios",",",,"Hellenic Military Academy"

http://dbpedia.org/resource/Alexandros_Papagos,"Alexandros Papagos",",",""

http://dbpedia.org/resource/Alexandros_Papanastasiou,"Alexandros Papanastasiou","Agricultural and Labour Party",""

http://dbpedia.org/resource/Alexandros_Sakellariou,"Alexandros Sakellariou",",",""

http://dbpedia.org/resource/Alexandros_Svolos,"Alexandros Svolos","Socialist Party of Greece, Union of People's Party",""

http://dbpedia.org/resource/Alexandros_Zaimis,"Alexandros Zaimis","National Party (Greece)",""

http://dbpedia.org/resource/Alexis_Tsipras,"Alexis Tsipras","Syriza",""

http://dbpedia.org/resource/Anastasios_Charalambis,"Anastasios Charalambis",",",""

http://dbpedia.org/resource/Andreas_Metaxas,"Andreas Metaxas","Russian Party",""

http://dbpedia.org/resource/Andreas_Michalakopoulos,"Andreas Michalakopoulos",",",""

http://dbpedia.org/resource/Andreas_Papandreou,"Andreas Papandreou","PASOK","Harvard University, National Technical University of Athens"

http://dbpedia.org/resource/Andreas_Zaimis,"Andreas Zaimis",",",""

http://dbpedia.org/resource/Antonios_Kriezis,"Antonios Kriezis",",",""

http://dbpedia.org/resource/Antonis_Samaras,"Antonis Samaras",",",""

http://dbpedia.org/resource/Aristeidis_Moraitinis,"Aristeidis Moraitinis",",",""

http://dbpedia.org/resource/Athanasios_Eftaxias,"Athanasios Eftaxias",",",""

http://dbpedia.org/resource/Athanasios_Miaoulis,"Athanasios Miaoulis",",",""

http://dbpedia.org/resource/Augustinos_Kapodistrias,"Augustinos Kapodistrias","Russian Party",""

http://dbpedia.org/resource/Benizelos_Roufos,"Benizelos Roufos",",",""

http://dbpedia.org/resource/Charilaos_Trikoupis,"Charilaos Trikoupis","New Party (Greece, 1873)",""

http://dbpedia.org/resource/Constantine_I_of_Greece, "Constantine I of Greece", "", ""

http://dbpedia.org/resource/Constantine_II_of_Greece, "Constantine II of Greece", "", ""

http://dbpedia.org/resource/Constantine_Kollias, "Constantine Kollias", "", ""

http://dbpedia.org/resource/Dimitrios_Gounaris, "Dimitrios Gounaris", "People's Party (Greece)", ""

http://dbpedia.org/resource/Dimitrios_Ioannidis, "Dimitrios Ioannidis", "", "Hellenic Military Academy"

http://dbpedia.org/resource/Dimitrios_Kiousopoulos, "Dimitrios Kiousopoulos", "", ""

http://dbpedia.org/resource/Dimitrios_Maximos, "Dimitrios Maximos", "People's Party (Greece)", "National and Kapodistrian University of Athens"

http://dbpedia.org/resource/Dimitrios_Partsalidis, "Dimitrios Partsalidis", "", ""

http://dbpedia.org/resource/Dimitrios_Rallis, "Dimitrios Rallis", "", "National and Kapodistrian University of Athens"

http://dbpedia.org/resource/Dimitrios_Valvis, "Dimitrios Valvis", "", ""

http://dbpedia.org/resource/Dimitrios_Voulgaris, "Dimitrios Voulgaris", "", ""

http://dbpedia.org/resource/Diomidis_Kyriakos, "Diomidis Kyriakos", "", ""

http://dbpedia.org/resource/Eleftherios_Venizelos, "Eleftherios Venizelos", "Liberal Party (Greece)", "National and Kapodistrian University of Athens"

http://dbpedia.org/resource/Emmanouil_Tsouderos, "Emmanouil Tsouderos", "", ""

http://dbpedia.org/resource/Epameinondas_Deligeorgis, "Epameinondas Deligeorgis", "", ""

http://dbpedia.org/resource/Evripidis_Bakirtzis, "Evripidis Bakirtzis", "", "Hellenic Military Academy"

http://dbpedia.org/resource/Gennaios_Kolokotronis, "Gennaios Kolokotronis", "Military", ""

http://dbpedia.org/resource/George_I_of_Greece, "George I of Greece", "", ""

http://dbpedia.org/resource/George_II_of_Greece, "George II of Greece", "", ""

http://dbpedia.org/resource/George_Papandreou, "George Papandreou", "", ""

http://dbpedia.org/resource/Georgios_Athanasiadis-Novas, "Georgios Athanasiadis-Novas", "Centre Union", ""

http://dbpedia.org/resource/Georgios_Kafantaris, "Georgios Kafantaris", "", ""

http://dbpedia.org/resource/Georgios_Kondylis, "Georgios Kondylis", "National Democratic Party (Greece)", ""

http://dbpedia.org/resource/Georgios_Kountouriotis, "Georgios Kountouriotis", "French Party", ""

http://dbpedia.org/resource/Georgios_Mavromichalis, "Georgios Mavromichalis", "", ""

http://dbpedia.org/resource/Georgios_Papadopoulos, "Georgios Papadopoulos", "National Political Union (1984)", ""

http://dbpedia.org/resource/Georgios_Papandreou, "Georgios Papandreou", "", ""

http://dbpedia.org/resource/Georgios_Rallis, "Georgios Rallis", "New Democracy (Greece)", "National and Kapodistrian University of Athens"

http://dbpedia.org/resource/Georgios_Theotokis, "Georgios Theotokis", "New Party (Greece, 1873)", ""

http://dbpedia.org/resource/Georgios_Tsolakoglou, "Georgios Tsolakoglou", "", ""

http://dbpedia.org/resource/Ignaz_von_Rudhart, "Ignaz von Rudhart", "", ""

http://dbpedia.org/resource/Ilias_Tsirimokos, "Ilias Tsirimokos", "", ""

http://dbpedia.org/resource/Ioannis_Grivas, "Ioannis Grivas", "Independent politician", "National and Kapodistrian University of Athens"

http://dbpedia.org/resource/Ioannis_Kapodistrias, "Ioannis Kapodistrias", "Russian Party", "University of Padua"

http://dbpedia.org/resource/Ioannis_Kolettis, "Ioannis Kolettis", "French Party", "University of Pisa"

http://dbpedia.org/resource/Ioannis_Metaxas, "Ioannis Metaxas", "Freethinkers' Party, Independent politician", "University of Athens"

http://dbpedia.org/resource/Ioannis_Paraskevopoulos, "Ioannis Paraskevopoulos", "", ""

http://dbpedia.org/resource/Ioannis_Rallis, "Ioannis Rallis", "", ""

http://dbpedia.org/resource/Ioannis_Theotokis, "Ioannis Theotokis", "", ""

http://dbpedia.org/resource/Josef_Ludwig_von_Armansperg, "Josef Ludwig von Armansperg", "", ""

http://dbpedia.org/resource/Kitsos_Tzavelas, "Kitsos Tzavelas", "Russian Party", ""

http://dbpedia.org/resource/Konstantinos_Demertzis, "Konstantinos Demertzis", "Freethinkers' Party", ""

http://dbpedia.org/resource/Konstantinos_Dovas, "Konstantinos Dovas", "", "Hellenic Military Academy"

http://dbpedia.org/resource/Konstantinos_Georgakopoulos, "Konstantinos Georgakopoulos", "People's Party (Greece)", ""

http://dbpedia.org/resource/Konstantinos_Kanaris, "Konstantinos Kanaris", "Russian Party", ""

http://dbpedia.org/resource/Konstantinos_Karamanlis, "Konstantinos Karamanlis", "", "National and Kapodistrian University of Athens"

http://dbpedia.org/resource/Konstantinos_Konstantopoulos, "Konstantinos Konstantopoulos", "", ""

http://dbpedia.org/resource/Konstantinos_Kotzias, "Konstantinos Kotzias", "", ""

http://dbpedia.org/resource/Konstantinos_Logothetopoulos, "Konstantinos Logothetopoulos", "", ""

http://dbpedia.org/resource/Konstantinos_Mitsotakis, "Konstantinos Mitsotakis", "Centre Union, Independent p

http://dbpedia.org/resource/Konstantinos_Tsaldaris, "Konstantinos Tsaldaris", "", ""

http://dbpedia.org/resource/Kostas_Karamanlis, "Kostas Karamanlis", "New Democracy (Greece)", "Fletcher Scho

http://dbpedia.org/resource/Kyriakos_Mitsotakis, "Kyriakos Mitsotakis", "New Democracy (Greece)", "Harvard Un

http://dbpedia.org/resource/Kyriakoulis_Mavromichalis, "Kyriakoulis Mavromichalis", "", ""

http://dbpedia.org/resource/Lucas_Papademos, "Lucas Papademos", "Independent politician", ""

http://dbpedia.org/resource/Markos_Vafeiadis, "Markos Vafeiadis", "Communist Party of Greece, PASOK", ""

http://dbpedia.org/resource/Nikolaos_Deligiannis, "Nikolaos Deligiannis", "", ""

http://dbpedia.org/resource/Nikolaos_Kalogeropoulos, "Nikolaos Kalogeropoulos", "", ""

http://dbpedia.org/resource/Nikolaos_Plastiras, "Nikolaos Plastiras", "National Progressive Center Union", ""

http://dbpedia.org/resource/Nikolaos_Stratos, "Nikolaos Stratos", "", ""

http://dbpedia.org/resource/Nikolaos_Triantafyllakos, "Nikolaos Triantafyllakos", "", ""

http://dbpedia.org/resource/Nikos_Zachariadis, "Nikos Zachariadis", "", ""

http://dbpedia.org/resource/Otto_of_Greece, "Otto of Greece", "", ""

http://dbpedia.org/resource/Panagiotis_Kanellopoulos, "Panagiotis Kanellopoulos", "National Radical Union", ""

http://dbpedia.org/resource/Panagiotis_Pikrammenos, "Panagiotis Pikrammenos", "Independent politician, New

http://dbpedia.org/resource/Panagiotis_Pipinelis, "Panagiotis Pipinelis", "Greek Rally, National Radical Union", ""

http://dbpedia.org/resource/Panagiotis_Poulitsas, "Panagiotis Poulitsas", "", ""

http://dbpedia.org/resource/Panagis_Tsaldaris, "Panagis Tsaldaris", "People's Party (Greece)", ""

http://dbpedia.org/resource/Petros_Protopapadakis, "Petros Protopapadakis", "National Party (Greece), People's

http://dbpedia.org/resource/Petros_Voulgaris, "Petros Voulgaris", "", ""

http://dbpedia.org/resource/Sotirios_Krokidas, "Sotirios Krokidas", "", ""

http://dbpedia.org/resource/Sotirios_Sotiropoulos, "Sotirios Sotiropoulos", "", ""

http://dbpedia.org/resource/Spyridon_Lambros, "Spyridon Lambros", "", ""

http://dbpedia.org/resource/Spyridon_Trikoupis, "Spyridon Trikoupis", "English Party", ""

http://dbpedia.org/resource/Spyros_Markezinis, "Spyros Markezinis", "New Party (Greece, 1947)", ""

http://dbpedia.org/resource/Stefanos_Stefanopoulos, "Stefanos Stefanopoulos", "Liberal Democratic Center", ""

http://dbpedia.org/resource/Stylios_Gonatas, "Stylios Gonatas", "", "Hellenic Military Academy"

http://dbpedia.org/resource/Stylios_Mavromichalis, "Stylios Mavromichalis", "", ""

http://dbpedia.org/resource/Themistoklis_Sofoulis, "Themistoklis Sofoulis", "", ""

http://dbpedia.org/resource/Theodoros_Kolokotronis, "Theodoros Kolokotronis", "", ""

http://dbpedia.org/resource/Theodoros_Negris, "Theodoros Negris", "", ""

http://dbpedia.org/resource/Theodoros_Pangalos, "Theodoros Pangalos", "", "Hellenic Military Academy"

http://dbpedia.org/resource/Thrasyvoulos_Zaimis, "Thrasyvoulos Zaimis", "", ""

http://dbpedia.org/resource/Tzannis_Tzannetakis, "Tzannis Tzannetakis", "New Democracy (Greece)", ""

http://dbpedia.org/resource/Vassiliki_Thanou-Christophilou, "Vassiliki Thanou-Christophilou", "Independent poli

http://dbpedia.org/resource/Xenophon_Zolotas, "Xenophon Zolotas", "Independent politician", "Leipzig Universit

http://dbpedia.org/resource/Zinovios_Valvis, "Zinovios Valvis", "Independent politician", ""

Exercise 2

1)

To retrieve the official name and population of every municipality (δήμος) in Greece as defined in the Kallikratis administrative geography dataset, we execute a SPARQL 1.1 query over the Kallikratis ontology and dataset. Municipalities are represented as instances of the class gag:Δήμος, while their official names and population values are stored using the data properties gag:έχει_επίσημο_όνομα and gag:έχει_πληθυσμό, respectively.

The following SPARQL query returns, for each municipality, its official name and its population:

```
PREFIX gag: <http://geo.linkedopendata.gr/gag/ontology/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT ?name ?population
WHERE {
  ?d rdf:type gag:Δήμος ;
  gag:έχει_επίσημο_όνομα ?name ;
  gag:έχει_πληθυσμό ?population .
}
ORDER BY ?name
```

This query identifies all resources classified as municipalities and retrieves the corresponding official name and population values for each one. The results provide a complete list of Greek municipalities together with their demographic information, as stored in the linked data representation of the Kallikratis plan.

name	population
ΔΗΜΟΣ ΑΒΔΗΡΩΝ	18573
ΔΗΜΟΣ ΑΓΑΘΟΝΗΣΙΟΥ	158
ΔΗΜΟΣ ΑΓΙΑΣ	14121
ΔΗΜΟΣ ΑΓΙΑΣ ΒΑΡΒΑΡΑΣ	30562
ΔΗΜΟΣ ΑΓΙΑΣ ΠΑΡΑΣΚΕΥΗΣ	56836
ΔΗΜΟΣ ΑΓΙΟΥ ΒΑΣΙΛΕΙΟΥ	10079
ΔΗΜΟΣ ΑΓΙΟΥ ΔΗΜΗΤΡΙΟΥ	65173
ΔΗΜΟΣ ΑΓΙΟΥ ΕΥΣΤΡΑΤΙΟΥ	371
ΔΗΜΟΣ ΑΓΙΟΥ ΝΙΚΟΛΑΟΥ	26227
ΔΗΜΟΣ ΑΓΙΩΝ ΑΝΑΡΓΥΡΩΝ - ΚΑΜΑΤΕΡΟΥ	55191
ΔΗΜΟΣ ΑΓΚΙΣΤΡΙΟΥ	920
ΔΗΜΟΣ ΑΓΡΑΦΩΝ	13604
ΔΗΜΟΣ ΑΓΡΙΝΙΟΥ	55110
ΔΗΜΟΣ ΑΘΗΝΑΙΩΝ	657701
ΔΗΜΟΣ ΑΙΓΑΛΕΩ	74046
ΔΗΜΟΣ ΑΙΓΙΑΛΕΙΑΣ	57993
ΔΗΜΟΣ ΑΙΓΙΝΑΣ	13552
ΔΗΜΟΣ ΑΚΤΙΟΥ-ΒΟΝΙΤΣΑΣ	17456
ΔΗΜΟΣ ΑΛΕΞΑΝΔΡΕΙΑΣ	28737
ΔΗΜΟΣ ΑΛΕΞΑΝΔΡΟΥΠΟΛΗΣ	65894
ΔΗΜΟΣ ΑΛΙΑΡΤΟΥ	9609
ΔΗΜΟΣ ΑΛΙΜΟΥ	38047

ΔΗΜΟΣ ΑΛΜΥΡΟΥ	21169
ΔΗΜΟΣ ΑΛΜΩΠΙΑΣ	27230
ΔΗΜΟΣ ΑΛΟΝΝΗΣΟΥ	27
ΔΗΜΟΣ ΑΜΑΡΙΟΥ	6215
ΔΗΜΟΣ ΑΜΑΡΟΥΣΙΟΥ	6947
ΔΗΜΟΣ ΑΜΟΡΓΟΥ	1859
ΔΗΜΟΣ ΑΜΠΕΛΟΚΗΠΩΝ-ΜΕΝΕΜΕΝΗΣ	42450
ΔΗΜΟΣ ΑΜΥΝΤΑΙΟΥ	17976
ΔΗΜΟΣ ΑΜΦΙΚΛΕΙΑΣ - ΕΛΑΤΕΙΑΣ	9679
ΔΗΜΟΣ ΑΜΦΙΛΟΧΙΑΣ	21445
ΔΗΜΟΣ ΑΜΦΙΠΟΛΗΣ	11185
ΔΗΜΟΣ ΑΝΑΤΟΛΙΚΗΣ ΜΑΝΗΣ	17157
ΔΗΜΟΣ ΑΝΑΦΗΣ	273
ΔΗΜΟΣ ΑΝΔΡΑΒΙΔΑΣ-ΚΥΛΛΗΝΗΣ	22364
ΔΗΜΟΣ ΑΝΔΡΙΤΣΑΙΝΑΣ-ΚΡΕΣΤΕΝΩΝ	21912
ΔΗΜΟΣ ΑΝΔΡΟΥ	10009
ΔΗΜΟΣ ΑΝΤΙΠΑΡΟΥ	430
ΔΗΜΟΣ ΑΝΩΓΕΙΩΝ	2507
ΔΗΜΟΣ ΑΠΟΚΟΡΩΝΟΥ	12703
ΔΗΜΟΣ ΑΡΓΙΘΕΑΣ	6084
ΔΗΜΟΣ ΑΡΓΟΥΣ-ΜΥΚΗΝΩΝ	23735
ΔΗΜΟΣ ΑΡΙΣΤΟΤΕΛΗ	18861
ΔΗΜΟΣ ΑΡΡΙΑΝΩΝ	17242
ΔΗΜΟΣ ΑΡΤΑΙΩΝ	40608
ΔΗΜΟΣ ΑΡΧΑΙΑΣ ΟΛΥΜΠΙΑΣ	18444
ΔΗΜΟΣ ΑΡΧΑΝΩΝ - ΑΣΤΕΡΟΥΣΙΩΝ	14503
ΔΗΜΟΣ ΑΣΠΡΟΠΥΡΓΟΥ	27741
ΔΗΜΟΣ ΑΣΤΥΠΑΛΛΙΑΣ	1238
ΔΗΜΟΣ ΑΧΑΡΝΩΝ	75819
ΔΗΜΟΣ ΒΑΡΗΣ - ΒΟΥΛΑΣ - ΒΟΥΛΙΑΓΜΕΝΗΣ	42972
ΔΗΜΟΣ ΒΕΛΟΥ-ΒΟΧΑΣ	18323
ΔΗΜΟΣ ΒΕΡΟΙΑΣ	63646
ΔΗΜΟΣ ΒΙΑΝΝΟΥ	6463
ΔΗΜΟΣ ΒΙΣΑΛΤΙΑΣ	26127
ΔΗΜΟΣ ΒΟΛΒΗΣ	26079
ΔΗΜΟΣ ΒΟΛΟΥ	141675
ΔΗΜΟΣ ΒΟΡΕΙΑΣ ΚΥΝΟΥΡΙΑΣ	12825
ΔΗΜΟΣ ΒΟΡΕΙΩΝ ΤΖΟΥΜΕΡΚΩΝ	7097
ΔΗΜΟΣ ΒΟΪΟΥ	22447
ΔΗΜΟΣ ΒΡΙΛΗΣΣΙΩΝ	25582
ΔΗΜΟΣ ΒΥΡΩΝΟΣ	61102
ΔΗΜΟΣ ΓΑΛΑΤΣΙΟΥ	58042
ΔΗΜΟΣ ΓΑΥΔΟΥ	98
ΔΗΜΟΣ ΓΕΩΡΓΙΟΥ ΚΑΡΑΪΣΚΑΚΗ	9016

ΔΗΜΟΣ ΓΛΥΦΑΔΑΣ	80409
ΔΗΜΟΣ ΓΟΡΤΥΝΑΣ	18264
ΔΗΜΟΣ ΓΟΡΤΥΝΙΑΣ	18327
ΔΗΜΟΣ ΓΡΕΒΕΝΩΝ	30564
ΔΗΜΟΣ ΔΑΦΝΗΣ - ΥΜΗΤΤΟΥ	34813
ΔΗΜΟΣ ΔΕΛΤΑ	30200
ΔΗΜΟΣ ΔΕΛΦΩΝ	31336
ΔΗΜΟΣ ΔΕΣΚΑΤΗΣ	7383
ΔΗΜΟΣ ΔΙΔΥΜΟΤΕΙΧΟΥ	23484
ΔΗΜΟΣ ΔΙΟΝΥΣΟΥ	31105
ΔΗΜΟΣ ΔΙΟΥ-ΟΛΥΜΠΟΥ	23857
ΔΗΜΟΣ ΔΙΡΦΥΩΝ-ΜΕΣΣΑΠΙΩΝ	21064
ΔΗΜΟΣ ΔΙΣΤΟΜΟΥ - ΑΡΑΧΟΒΑΣ - ΑΝΤΙΚΥΡΑΣ	4755
ΔΗΜΟΣ ΔΟΜΟΚΟΥ	14793
ΔΗΜΟΣ ΔΟΞΑΤΟΥ	17481
ΔΗΜΟΣ ΔΡΑΜΑΣ	56062
ΔΗΜΟΣ ΔΥΤΙΚΗΣ ΑΧΑΪΑΣ	28422
ΔΗΜΟΣ ΔΥΤΙΚΗΣ ΜΑΝΗΣ	8647
ΔΗΜΟΣ ΔΩΔΩΝΗΣ	13939
ΔΗΜΟΣ ΔΩΡΙΔΟΣ	16021
ΔΗΜΟΣ ΕΔΕΣΣΑΣ	29799
ΔΗΜΟΣ ΕΛΑΣΣΟΝΑΣ	37264
ΔΗΜΟΣ ΕΛΑΦΟΝΗΣΟΥ	745
ΔΗΜΟΣ ΕΛΕΥΣΙΝΑΣ	29868
ΔΗΜΟΣ ΕΛΛΗΝΙΚΟΥ - ΑΡΓΥΡΟΥΠΟΛΗΣ	34832
ΔΗΜΟΣ ΕΜΜΑΝΟΥΗΛ ΠΑΠΠΑ	18035
ΔΗΜΟΣ ΕΟΡΔΑΙΑΣ	44497
ΔΗΜΟΣ ΕΠΙΔΑΥΡΟΥ	9275
ΔΗΜΟΣ ΕΡΕΤΡΙΑΣ	13325
ΔΗΜΟΣ ΕΡΜΙΟΝΙΔΑΣ	12624
ΔΗΜΟΣ ΕΡΥΜΑΝΘΟΥ	13162
ΔΗΜΟΣ ΕΥΡΩΤΑ	18345
ΔΗΜΟΣ ΖΑΓΟΡΑΣ-ΜΟΥΡΕΣΙΟΥ	6936
ΔΗΜΟΣ ΖΑΓΟΡΙΟΥ	6032
ΔΗΜΟΣ ΖΑΚΥΝΘΟΥ	39015
ΔΗΜΟΣ ΖΑΧΑΡΩΣ	15409
ΔΗΜΟΣ ΖΗΡΟΥ	16494
ΔΗΜΟΣ ΖΙΤΣΑΣ	17293
ΔΗΜΟΣ ΖΩΓΡΑΦΟΥ	76115
ΔΗΜΟΣ ΗΓΟΥΜΕΝΙΤΣΗΣ	24692
ΔΗΜΟΣ ΗΛΙΔΑΣ	19723
ΔΗΜΟΣ ΗΛΙΟΥΠΟΛΕΩΣ	75904
ΔΗΜΟΣ ΗΡΑΚΛΕΙΑΣ	22021
ΔΗΜΟΣ ΗΡΑΚΛΕΙΟΥ	45926

ΔΗΜΟΣ ΗΡΑΚΛΕΙΟΥ	132361
ΔΗΜΟΣ ΘΑΣΟΥ	10939
ΔΗΜΟΣ ΘΕΡΜΑΪΚΟΥ	34686
ΔΗΜΟΣ ΘΕΡΜΗΣ	22659
ΔΗΜΟΣ ΘΕΡΜΟΥ	9299
ΔΗΜΟΣ ΘΕΣΣΑΛΟΝΙΚΗΣ	375276
ΔΗΜΟΣ ΘΗΒΑΙΩΝ	37524
ΔΗΜΟΣ ΘΗΡΑΣ	11051
ΔΗΜΟΣ ΙΑΣΜΟΥ	14890
ΔΗΜΟΣ ΙΕΡΑΠΕΤΡΑΣ	27911
ΔΗΜΟΣ ΙΕΡΑΣ ΠΟΛΗΣ ΜΕΣΟΛΟΓΓΙΟΥ	32830
ΔΗΜΟΣ ΙΗΤΩΝ	1838
ΔΗΜΟΣ ΙΘΑΚΗΣ	3084
ΔΗΜΟΣ ΙΚΑΡΙΑΣ	8312
ΔΗΜΟΣ ΙΛΙΟΥ (ΝΕΩΝ ΛΙΟΣΙΩΝ)	80859
ΔΗΜΟΣ ΙΣΤΙΑΙΑΣ-ΑΙΔΗΨΟΥ	17475
ΔΗΜΟΣ ΙΩΑΝΝΙΤΩΝ	97657
ΔΗΜΟΣ ΚΑΒΑΛΑΣ	74120
ΔΗΜΟΣ ΚΑΙΣΑΡΙΑΝΗΣ	26419
ΔΗΜΟΣ ΚΑΛΑΒΡΥΤΩΝ	17602
ΔΗΜΟΣ ΚΑΛΑΜΑΡΙΑΣ	87255
ΔΗΜΟΣ ΚΑΛΑΜΑΤΑΣ	19418
ΔΗΜΟΣ ΚΑΛΑΜΠΑΚΑΣ	27555
ΔΗΜΟΣ ΚΑΛΛΙΘΕΑΣ	109609
ΔΗΜΟΣ ΚΑΛΥΜΝΙΩΝ	16441
ΔΗΜΟΣ ΚΑΝΤΑΝΟΥ - ΣΕΛΙΝΟΥ	7334
ΔΗΜΟΣ ΚΑΡΔΙΤΣΑΣ	54681
ΔΗΜΟΣ ΚΑΡΠΑΘΟΥ	6511
ΔΗΜΟΣ ΚΑΡΠΕΝΗΣΙΟΥ	18449
ΔΗΜΟΣ ΚΑΡΥΣΤΟΥ	14645
ΔΗΜΟΣ ΚΑΣΟΥ	990
ΔΗΜΟΣ ΚΑΣΣΑΝΔΡΑΣ	16153
ΔΗΜΟΣ ΚΑΣΤΟΡΙΑΣ	35333
ΔΗΜΟΣ ΚΑΤΕΡΙΝΗΣ	83764
ΔΗΜΟΣ ΚΑΤΩ ΝΕΥΡΟΚΟΠΙΟΥ	6955
ΔΗΜΟΣ ΚΕΑΣ	2417
ΔΗΜΟΣ ΚΕΝΤΡΙΚΩΝ ΤΖΟΥΜΕΡΚΩΝ	12069
ΔΗΜΟΣ ΚΕΡΑΤΣΙΝΙΟΥ - ΔΡΑΠΕΤΣΩΝΑΣ	89046
ΔΗΜΟΣ ΚΕΡΚΥΡΑΣ	106837
ΔΗΜΟΣ ΚΕΦΑΛΟΝΙΑΣ	32858
ΔΗΜΟΣ ΚΗΦΙΣΙΑΣ	59887
ΔΗΜΟΣ ΚΙΛΕΛΕΡ	23213
ΔΗΜΟΣ ΚΙΛΚΙΣ	56336
ΔΗΜΟΣ ΚΙΜΩΛΟΥ	769

ΔΗΜΟΣ ΚΙΣΣΑΜΟΥ	11820
ΔΗΜΟΣ ΚΟΖΑΝΗΣ	68680
ΔΗΜΟΣ ΚΟΜΟΤΗΝΗΣ	60574
ΔΗΜΟΣ ΚΟΝΙΤΣΑΣ	9294
ΔΗΜΟΣ ΚΟΡΔΕΛΙΟΥ-ΕΥΟΣΜΟΥ	54787
ΔΗΜΟΣ ΚΟΡΙΝΘΙΩΝ	55258
ΔΗΜΟΣ ΚΟΡΥΔΑΛΛΟΥ	67456
ΔΗΜΟΣ ΚΡΩΠΙΑΣ	25325
ΔΗΜΟΣ ΚΥΘΗΡΩΝ	3398
ΔΗΜΟΣ ΚΥΘΝΟΥ	1608
ΔΗΜΟΣ ΚΥΜΗΣ-ΑΛΙΒΕΡΙΟΥ	32339
ΔΗΜΟΣ ΚΩ	14846
ΔΗΜΟΣ ΛΑΓΚΑΔΑ	37528
ΔΗΜΟΣ ΛΑΜΙΕΩΝ	71693
ΔΗΜΟΣ ΛΑΡΙΣΑΙΩΝ	138242
ΔΗΜΟΣ ΛΑΥΡΕΩΤΙΚΗΣ	24545
ΔΗΜΟΣ ΛΕΒΑΔΕΩΝ	30974
ΔΗΜΟΣ ΛΕΙΨΩΝ	698
ΔΗΜΟΣ ΛΕΡΟΥ	8207
ΔΗΜΟΣ ΛΕΣΒΟΥ	84946
ΔΗΜΟΣ ΛΕΥΚΑΔΑΣ	21414
ΔΗΜΟΣ ΛΗΜΝΟΥ	18104
ΔΗΜΟΣ ΛΙΜΝΗΣ ΠΛΑΣΤΗΡΑ	7392
ΔΗΜΟΣ ΛΟΚΡΩΝ	24611
ΔΗΜΟΣ ΛΟΥΤΡΑΚΙΟΥ-ΑΓ.ΘΕΟΔΩΡΩΝ	17116
ΔΗΜΟΣ ΛΥΚΟΒΡΥΣΗΣ - ΠΕΥΚΗΣ	28003
ΔΗΜΟΣ ΜΑΚΡΑΚΩΜΗΣ	21570
ΔΗΜΟΣ ΜΑΛΕΒΙΖΙΟΥ	20159
ΔΗΜΟΣ ΜΑΝΔΡΑΣ - ΕΙΔΥΛΛΙΑΣ	20098
ΔΗΜΟΣ ΜΑΝΤΟΥΔΙΟΥ-ΛΙΜΝΗΣ-ΑΓΙΑΣ ΑΝΝΑΣ	15049
ΔΗΜΟΣ ΜΑΡΑΘΩΝΟΣ	26899
ΔΗΜΟΣ ΜΑΡΚΟΠΟΥΛΟΥ ΜΕΣΟΓΑΙΑΣ	15608
ΔΗΜΟΣ ΜΑΡΩΝΕΙΑΣ-ΣΑΠΩΝ	17186
ΔΗΜΟΣ ΜΕΓΑΛΟΠΟΛΗΣ	13152
ΔΗΜΟΣ ΜΕΓΑΝΗΣΙΟΥ	1092
ΔΗΜΟΣ ΜΕΓΑΡΕΩΝ	28943
ΔΗΜΟΣ ΜΕΓΙΣΤΗΣ	244
ΔΗΜΟΣ ΜΕΣΣΗΝΗΣ	27521
ΔΗΜΟΣ ΜΕΤΑΜΟΡΦΩΣΕΩΣ	26448
ΔΗΜΟΣ ΜΕΤΣΟΒΟΥ	7835
ΔΗΜΟΣ ΜΗΛΟΥ	4771
ΔΗΜΟΣ ΜΙΝΩΑ ΠΕΔΙΑΔΑΣ	20332
ΔΗΜΟΣ ΜΟΝΕΜΒΑΣΙΑΣ	23853
ΔΗΜΟΣ ΜΟΣΧΑΤΟΥ - ΤΑΥΡΟΥ	38116

ΔΗΜΟΣ ΜΟΥΖΑΚΙΟΥ	14121
ΔΗΜΟΣ ΜΥΚΗΣ	15724
ΔΗΜΟΣ ΜΥΚΟΝΟΥ	9320
ΔΗΜΟΣ ΜΥΛΟΠΟΤΑΜΟΥ	15092
ΔΗΜΟΣ ΜΩΛΟΥ - ΑΓΙΟΥ ΚΩΝΣΤΑΝΤΙΝΟΥ	15142
ΔΗΜΟΣ ΝΑΞΟΥ ΜΙΚΡΩΝ ΚΥΚΛΑΔΩΝ	19074
ΔΗΜΟΣ ΝΑΟΥΣΑΣ	33334
ΔΗΜΟΣ ΝΑΥΠΑΚΤΙΑΣ	29842
ΔΗΜΟΣ ΝΑΥΠΛΙΕΩΝ	33406
ΔΗΜΟΣ ΝΕΑΠΟΛΗΣ-ΣΥΚΕΩΝ	86417
ΔΗΜΟΣ ΝΕΑΣ ΖΙΧΝΗΣ	13579
ΔΗΜΟΣ ΝΕΑΣ ΙΩΝΙΑΣ	66017
ΔΗΜΟΣ ΝΕΑΣ ΠΡΟΠΟΝΤΙΔΑΣ	32433
ΔΗΜΟΣ ΝΕΑΣ ΣΜΥΡΝΗΣ	73986
ΔΗΜΟΣ ΝΕΜΕΑΣ	7774
ΔΗΜΟΣ ΝΕΣΤΟΡΙΟΥ	3542
ΔΗΜΟΣ ΝΕΣΤΟΥ	23486
ΔΗΜΟΣ ΝΙΚΑΙΑΣ - ΑΓΙΟΥ Ι. ΡΕΝΤΗ	94592
ΔΗΜΟΣ ΝΙΚΟΛΑΟΥ ΣΚΟΥΦΑ	15235
ΔΗΜΟΣ ΝΙΣΥΡΟΥ	948
ΔΗΜΟΣ ΝΟΤΙΑΣ ΚΥΝΟΥΡΙΑΣ	9001
ΔΗΜΟΣ ΝΟΤΙΟΥ ΠΗΛΙΟΥ	9925
ΔΗΜΟΣ ΞΑΝΘΗΣ	52147
ΔΗΜΟΣ ΞΗΡΟΜΕΡΟΥ	13717
ΔΗΜΟΣ ΞΥΛΟΚΑΣΤΡΟΥ-ΕΥΡΩΣΤΙΝΗΣ	20084
ΔΗΜΟΣ ΟΙΝΟΥΣΣΩΝ	105
ΔΗΜΟΣ ΟΙΧΑΛΙΑΣ	14987
ΔΗΜΟΣ ΟΡΕΣΤΙΑΔΟΣ	39485
ΔΗΜΟΣ ΟΡΕΣΤΙΔΟΣ	6571
ΔΗΜΟΣ ΟΡΟΠΕΔΙΟΥ ΛΑΣΙΘΙΟΥ	3152
ΔΗΜΟΣ ΟΡΧΟΜΕΝΟΥ	6123
ΔΗΜΟΣ ΠΑΓΓΑΙΟΥ	33683
ΔΗΜΟΣ ΠΑΙΑΝΙΑΣ	19636
ΔΗΜΟΣ ΠΑΙΟΝΙΑΣ	32720
ΔΗΜΟΣ ΠΑΛΑΙΟΥ ΦΑΛΗΡΟΥ	64759
ΔΗΜΟΣ ΠΑΛΑΜΑ	19144
ΔΗΜΟΣ ΠΑΛΛΗΝΗΣ	33624
ΔΗΜΟΣ ΠΑΞΩΝ	2438
ΔΗΜΟΣ ΠΑΠΑΓΟΥ - ΧΟΛΑΡΓΟΥ	45373
ΔΗΜΟΣ ΠΑΡΑΝΕΣΤΙΟΥ	5925
ΔΗΜΟΣ ΠΑΡΓΑΣ	12944
ΔΗΜΟΣ ΠΑΡΟΥ	10760
ΔΗΜΟΣ ΠΑΤΜΟΥ	3044
ΔΗΜΟΣ ΠΑΤΡΕΩΝ	202757

ΔΗΜΟΣ ΠΑΥΛΟΥ ΜΕΛΑ	84397
ΔΗΜΟΣ ΠΕΙΡΑΙΩΣ	175697
ΔΗΜΟΣ ΠΕΛΛΑΣ	60601
ΔΗΜΟΣ ΠΕΝΤΕΛΗΣ	30511
ΔΗΜΟΣ ΠΕΡΑΜΑΤΟΣ	2572
ΔΗΜΟΣ ΠΕΡΙΣΤΕΡΙΟΥ	137918
ΔΗΜΟΣ ΠΕΤΡΟΥΠΟΛΕΩΣ	48327
ΔΗΜΟΣ ΠΗΝΕΙΟΥ	20232
ΔΗΜΟΣ ΠΛΑΤΑΝΙΑ	18622
ΔΗΜΟΣ ΠΟΛΥΓΥΡΟΥ	22189
ΔΗΜΟΣ ΠΟΡΟΥ	4348
ΔΗΜΟΣ ΠΡΕΒΕΖΗΣ	29918
ΔΗΜΟΣ ΠΡΕΣΠΩΝ	2511
ΔΗΜΟΣ ΠΡΟΣΟΤΣΑΝΗΣ	16481
ΔΗΜΟΣ ΠΥΔΝΑΣ-ΚΟΛΙΝΔΡΟΥ	18445
ΔΗΜΟΣ ΠΥΛΑΙΑΣ-ΧΟΡΤΙΑΤΗ	46904
ΔΗΜΟΣ ΠΥΛΗΣ	20019
ΔΗΜΟΣ ΠΥΛΟΥ-ΝΕΣΤΟΡΟΣ	20315
ΔΗΜΟΣ ΠΥΡΓΟΥ	51777
ΔΗΜΟΣ ΡΩΓΩΝΙΟΥ	9157
ΔΗΜΟΣ ΡΑΦΗΝΑΣ - ΠΙΚΕΡΜΙΟΥ	14840
ΔΗΜΟΣ ΡΕΘΥΜΝΗΣ	46558
ΔΗΜΟΣ ΡΗΓΑ ΦΕΡΡΑΙΟΥ	12096
ΔΗΜΟΣ ΡΟΔΟΥ	120312
ΔΗΜΟΣ ΣΑΛΑΜΙΝΟΣ	38022
ΔΗΜΟΣ ΣΑΜΟΘΡΑΚΗΣ	2723
ΔΗΜΟΣ ΣΑΜΟΥ	32050
ΔΗΜΟΣ ΣΑΡΩΝΙΚΟΥ	26320
ΔΗΜΟΣ ΣΕΡΒΙΩΝ-ΒΕΛΒΕΝΤΟΥ	17657
ΔΗΜΟΣ ΣΕΡΙΦΟΥ	1414
ΔΗΜΟΣ ΣΕΡΡΩΝ	73946
ΔΗΜΟΣ ΣΗΤΕΙΑΣ	19029
ΔΗΜΟΣ ΣΙΘΩΝΙΑΣ	12927
ΔΗΜΟΣ ΣΙΚΙΝΟΥ	238
ΔΗΜΟΣ ΣΙΚΥΩΝΙΩΝ	24666
ΔΗΜΟΣ ΣΙΝΤΙΚΗΣ	27383
ΔΗΜΟΣ ΣΙΦΝΟΥ	2442
ΔΗΜΟΣ ΣΚΙΑΘΟΥ	616
ΔΗΜΟΣ ΣΚΟΠΕΛΟΥ	4696
ΔΗΜΟΣ ΣΚΥΔΡΑΣ	19986
ΔΗΜΟΣ ΣΚΥΡΟΥ	2602
ΔΗΜΟΣ ΣΟΥΛΙΟΥ	10951
ΔΗΜΟΣ ΣΟΥΦΛΙΟΥ	15338
ΔΗΜΟΣ ΣΟΦΑΔΩΝ	23043

ΔΗΜΟΣ ΣΠΑΡΤΗΣ	38079
ΔΗΜΟΣ ΣΠΑΤΩΝ - ΑΡΤΕΜΙΔΟΣ	27594
ΔΗΜΟΣ ΣΠΕΤΣΩΝ	3916
ΔΗΜΟΣ ΣΤΥΛΙΔΟΣ	14686
ΔΗΜΟΣ ΣΥΜΗΣ	910
ΔΗΜΟΣ ΣΥΡΟΥ-ΕΡΜΟΥΠΟΛΗΣ	19782
ΔΗΜΟΣ ΣΦΑΚΙΩΝ	2446
ΔΗΜΟΣ ΤΑΝΑΓΡΑΣ	21652
ΔΗΜΟΣ ΤΕΜΠΩΝ	16466
ΔΗΜΟΣ ΤΗΛΟΥ	1139
ΔΗΜΟΣ ΤΗΝΟΥ	8574
ΔΗΜΟΣ ΤΟΠΕΙΡΟΥ	10867
ΔΗΜΟΣ ΤΡΙΚΚΑΙΩΝ	73350
ΔΗΜΟΣ ΤΡΙΠΟΛΗΣ	23665
ΔΗΜΟΣ ΤΡΙΦΥΛΙΑΣ	33581
ΔΗΜΟΣ ΤΡΟΙΖΗΝΙΑΣ	7376
ΔΗΜΟΣ ΤΥΡΝΑΒΟΥ	19979
ΔΗΜΟΣ ΥΔΡΑΣ	2719
ΔΗΜΟΣ ΦΑΙΣΤΟΥ	24228
ΔΗΜΟΣ ΦΑΡΚΑΔΟΝΑΣ	15881
ΔΗΜΟΣ ΦΑΡΣΑΛΩΝ	14648
ΔΗΜΟΣ ΦΙΛΑΔΕΛΦΕΙΑΣ - ΧΑΛΚΗΔΟΝΟΣ	34224
ΔΗΜΟΣ ΦΙΛΙΑΤΩΝ	10448
ΔΗΜΟΣ ΦΙΛΟΘΕΗΣ - ΨΥΧΙΚΟΥ	22480
ΔΗΜΟΣ ΦΛΩΡΙΝΑΣ	33282
ΔΗΜΟΣ ΦΟΛΕΓΑΝΔΡΟΥ	667
ΔΗΜΟΣ ΦΟΥΡΝΩΝ ΚΟΡΣΕΩΝ	1469
ΔΗΜΟΣ ΦΥΛΗΣ	30256
ΔΗΜΟΣ ΧΑΛΑΝΔΡΙΟΥ	71684
ΔΗΜΟΣ ΧΑΛΚΗΔΟΝΟΣ	35145
ΔΗΜΟΣ ΧΑΛΚΗΣ	984
ΔΗΜΟΣ ΧΑΛΚΙΔΕΩΝ	92202
ΔΗΜΟΣ ΧΑΝΙΩΝ	95456
ΔΗΜΟΣ ΧΑΪΔΑΡΙΟΥ	46276
ΔΗΜΟΣ ΧΕΡΣΟΝΗΣΟΥ	22708
ΔΗΜΟΣ ΧΙΟΥ	51936
ΔΗΜΟΣ ΨΑΡΩΝ	422
ΔΗΜΟΣ ΩΡΑΙΟΚΑΣΤΡΟΥ	22456
ΔΗΜΟΣ ΩΡΩΠΟΥ	33448

2)

To retrieve the full hierarchical structure of Greek administrative divisions according to the Kallikratis plan, we use SPARQL to extract three levels of information: regions (Περιφέρειες), regional units (Περιφερειακές Ενότητες), and municipalities (Δήμοι).

The ontology encodes these units using the classes `gag:Περιφέρεια`, `gag:Περιφερειακή_Ενότητα`, and `gag:Δήμος`, respectively.

Administrative containment is modeled with the object property `gag:ανήκει_σε` ("belongs to"), and official names are provided by the datatype property `gag:έχει_επίσημο_όνομα`.

The following SPARQL query retrieves, for each region, its official name, along with the official names of all regional units that belong to it and the official names of all municipalities that belong to each regional unit:

```
PREFIX gag: <http://geo.linkedopendata.gr/gag/ontology/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT ?regionName ?unitName ?municipalityName
WHERE {
  ?region rdf:type gag:Περιφέρεια ;
    gag:έχει_επίσημο_όνομα ?regionName .

  ?unit rdf:type gag:Περιφερειακή_Ενότητα ;
    gag:ανήκει_σε ?region ;
    gag:έχει_επίσημο_όνομα ?unitName .

  ?municipality rdf:type gag:Δήμος ;
    gag:ανήκει_σε ?unit ;
    gag:έχει_επίσημο_όνομα ?municipalityName .
}
ORDER BY ?regionName ?unitName ?municipalityName
```

This query returns a structured list organized hierarchically by **Region → Regional Unit → Municipality**, fully satisfying the requirements of the assignment.

Column1	Column2	Column3
regionName	unitName	municipali
ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΔΡΑΜΑΣ	ΔΗΜΟΣ ΔΡΑΜΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΔΡΑΜΑΣ	ΔΗΜΟΣ ΔΡΑΜΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΔΡΑΜΑΣ	ΔΗΜΟΣ ΚΑΡΔΙΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΔΡΑΜΑΣ	ΔΗΜΟΣ ΠΑΡΑΝΕΣΤΙΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΔΡΑΜΑΣ	ΔΗΜΟΣ ΠΙΣΣΙΝΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΕΒΡΟΥ	ΔΗΜΟΣ ΑΓ. ΚΩΣΤΑΝΤΙΝΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΕΒΡΟΥ	ΔΗΜΟΣ ΔΙΦΛΩ
ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΕΒΡΟΥ	ΔΗΜΟΣ ΟΡΕΙΝΗΣ
ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΕΒΡΟΥ	ΔΗΜΟΣ ΣΑΡΑΝΤΑ
ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΕΒΡΟΥ	ΔΗΜΟΣ ΣΟΦΙΑΔΩΝ
ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΘΑΣΟΥ	ΔΗΜΟΣ ΘΑΣΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΑΒΑΛΑΣ	ΔΗΜΟΣ ΚΑΒΑΛΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΑΒΑΛΑΣ	ΔΗΜΟΣ ΝΕΣΤΟΥ

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

ΠΕΡΙΦΕΡΕΙΑ ΣΤΕΡΕΑΣ ΕΛΛΑΔΑΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΦΘΙΩΤΙΔΑΣ	ΔΗΜΟΣ Μ
ΠΕΡΙΦΕΡΕΙΑ ΣΤΕΡΕΑΣ ΕΛΛΑΔΑΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΦΘΙΩΤΙΔΑΣ	ΔΗΜΟΣ ΣΤ
ΠΕΡΙΦΕΡΕΙΑ ΣΤΕΡΕΑΣ ΕΛΛΑΔΑΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΦΩΚΙΔΑΣ	ΔΗΜΟΣ Δ
ΠΕΡΙΦΕΡΕΙΑ ΣΤΕΡΕΑΣ ΕΛΛΑΔΑΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΦΩΚΙΔΑΣ	ΔΗΜΟΣ Δ

3)

To retrieve all municipalities of the Region of Peloponnese with more than 5,000 inhabitants, together with their official names, population values, and the regional unit to which they belong, we query the Kallikratis ontology and dataset.

The administrative hierarchy is modeled through the classes `gag:Περιφέρεια`, `gag:Περιφερειακή_Ενότητα`, and `gag:Δήμος`, with hierarchical containment expressed via the object property `gag:ανήκει_σε`.

Official names and population values are provided by the datatype properties `gag:έχει_επίσημο_όνομα` and `gag:έχει_πληθυσμό`.

The following SPARQL query returns the required information, organized by municipality and regional unit:

```
PREFIX gag: <http://geo.linkedopendata.gr/gag/ontology/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

SELECT ?municipalityName ?population ?unitName
WHERE {
  # Regions – pick only the Region of Peloponnese
  ?region rdf:type gag:Περιφέρεια ;
    gag:έχει_επίσημο_όνομα ?regionName .

  FILTER( CONTAINS( STR(?regionName), "ΠΕΛΟΠΟΝΝΗΣΟΥ" ) )

  # Regional units of Peloponnese
  ?unit rdf:type gag:Περιφερειακή_Ενότητα ;
    gag:ανήκει_σε ?region ;
    gag:έχει_επίσημο_όνομα ?unitName .

  # Municipalities belonging to each regional unit
  ?municipality rdf:type gag:Δήμος ;
    gag:ανήκει_σε ?unit ;
    gag:έχει_επίσημο_όνομα ?municipalityName ;
    gag:έχει_πληθυσμό ?population .

  # Population filter > 5000
  FILTER( xsd:integer(?population) > 5000 )
}
ORDER BY ?municipalityName ?unitName
```

This query produces a complete list of Peloponnese municipalities with population over 5,000, grouped by municipality and associated regional unit.

municipalityName	population	unitName
ΔΗΜΟΣ ΑΝΑΤΟΛΙΚΗΣ ΜΑΝΗΣ	17157	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΑΚΩΝΙΑΣ
ΔΗΜΟΣ ΑΡΓΟΥΣ-ΜΥΚΗΝΩΝ	23735	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΓΟΛΙΔΑΣ
ΔΗΜΟΣ ΒΕΛΟΥ-ΒΟΧΑΣ	18323	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΟΡΙΝΘΙΑΣ
ΔΗΜΟΣ ΒΟΡΕΙΑΣ ΚΥΝΟΥΡΙΑΣ	12825	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΚΑΔΙΑΣ
ΔΗΜΟΣ ΓΟΡΤΥΝΙΑΣ	18327	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΚΑΔΙΑΣ
ΔΗΜΟΣ ΔΥΤΙΚΗΣ ΜΑΝΗΣ	8647	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΕΣΣΗΝΙΑΣ
ΔΗΜΟΣ ΕΠΙΔΑΥΡΟΥ	9275	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΓΟΛΙΔΑΣ
ΔΗΜΟΣ ΕΡΜΙΟΝΙΔΑΣ	12624	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΓΟΛΙΔΑΣ
ΔΗΜΟΣ ΕΥΡΩΤΑ	18345	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΑΚΩΝΙΑΣ
ΔΗΜΟΣ ΚΑΛΑΜΑΤΑΣ	19418	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΕΣΣΗΝΙΑΣ
ΔΗΜΟΣ ΚΟΡΙΝΘΙΩΝ	55258	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΟΡΙΝΘΙΑΣ
ΔΗΜΟΣ ΛΟΥΤΡΑΚΙΟΥ-ΑΓ.ΘΕΟΔΩΡΩΝ	17116	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΟΡΙΝΘΙΑΣ
ΔΗΜΟΣ ΜΕΓΑΛΟΠΟΛΗΣ	13152	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΚΑΔΙΑΣ
ΔΗΜΟΣ ΜΕΣΣΗΝΗΣ	27521	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΕΣΣΗΝΙΑΣ
ΔΗΜΟΣ ΜΟΝΕΜΒΑΣΙΑΣ	23853	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΑΚΩΝΙΑΣ
ΔΗΜΟΣ ΝΑΥΠΛΙΕΩΝ	33406	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΓΟΛΙΔΑΣ
ΔΗΜΟΣ ΝΕΜΕΑΣ	7774	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΟΡΙΝΘΙΑΣ
ΔΗΜΟΣ ΝΟΤΙΑΣ ΚΥΝΟΥΡΙΑΣ	9001	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΚΑΔΙΑΣ
ΔΗΜΟΣ ΞΥΛΟΚΑΣΤΡΟΥ-ΕΥΡΩΣΤΙΝΗΣ	20084	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΟΡΙΝΘΙΑΣ
ΔΗΜΟΣ ΟΙΧΑΛΙΑΣ	14987	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΕΣΣΗΝΙΑΣ
ΔΗΜΟΣ ΠΥΛΟΥ-ΝΕΣΤΟΡΟΣ	20315	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΕΣΣΗΝΙΑΣ
ΔΗΜΟΣ ΣΙΚΥΩΝΙΩΝ	24666	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΟΡΙΝΘΙΑΣ
ΔΗΜΟΣ ΣΠΑΡΤΗΣ	38079	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΑΚΩΝΙΑΣ
ΔΗΜΟΣ ΤΡΙΠΟΛΗΣ	23665	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΚΑΔΙΑΣ
ΔΗΜΟΣ ΤΡΙΦΥΛΙΑΣ	33581	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΕΣΣΗΝΙΑΣ

4)

To determine which municipalities of the Region of Peloponnese lack seat (έδρα) information, we query the Kallikratis dataset using the classes gag:Περιφέρεια, gag:Περιφερειακή_Ενότητα, and gag:Δήμος.

Administrative containment is modeled using gag:ανήκει_σε, while official names and seat information are provided by the datatype properties gag:έχει_επίσημο_όνομα and gag:έχει_έδρα.

The following SPARQL query returns all municipalities in the Region of Peloponnese for which no seat information is available in the dataset:

```
PREFIX gag: <http://geo.linkedopendata.gr/gag/ontology/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT ?municipalityName
WHERE {
  ?region rdf:type gag:Περιφέρεια ;
    gag:έχει_επίσημο_όνομα ?regionName .
  FILTER(CONTAINS(STR(?regionName), "ΠΕΛΟΠΟΝΝΗΣΟΥ"))

  ?unit rdf:type gag:Περιφερειακή_Ενότητα ;
```

<pre> gag:ανήκει_σε ?region . ?municipality rdf:type gag:Δήμος ; gag:ανήκει_σε ?unit ; gag:έχει_επίσημο_όνομα ?municipalityName . FILTER NOT EXISTS { ?municipality gag:έχει_έδρα ?seat } } ORDER BY ?municipalityName </pre>

This query identifies all Peloponnese municipalities for which the dataset does not provide any seat information, and returns only their official names, as required.

municipalityName
ΔΗΜΟΣ ΑΝΑΤΟΛΙΚΗΣ ΜΑΝΗΣ
ΔΗΜΟΣ ΑΡΓΟΥΣ-ΜΥΚΗΝΩΝ
ΔΗΜΟΣ ΒΕΛΟΥ-ΒΟΧΑΣ
ΔΗΜΟΣ ΒΟΡΕΙΑΣ ΚΥΝΟΥΡΙΑΣ
ΔΗΜΟΣ ΓΟΡΤΥΝΙΑΣ
ΔΗΜΟΣ ΔΥΤΙΚΗΣ ΜΑΝΗΣ
ΔΗΜΟΣ ΕΛΑΦΟΝΗΣΟΥ
ΔΗΜΟΣ ΕΠΙΔΑΥΡΟΥ
ΔΗΜΟΣ ΕΡΜΙΟΝΙΔΑΣ
ΔΗΜΟΣ ΕΥΡΩΤΑ
ΔΗΜΟΣ ΚΑΛΑΜΑΤΑΣ
ΔΗΜΟΣ ΚΟΡΙΝΘΙΩΝ
ΔΗΜΟΣ ΛΟΥΤΡΑΚΙΟΥ-ΑΓ.ΘΕΟΔΩΡΩΝ
ΔΗΜΟΣ ΜΕΓΑΛΟΠΟΛΗΣ
ΔΗΜΟΣ ΜΕΣΣΗΝΗΣ
ΔΗΜΟΣ ΜΟΝΕΜΒΑΣΙΑΣ
ΔΗΜΟΣ ΝΑΥΠΛΙΕΩΝ
ΔΗΜΟΣ ΝΕΜΕΑΣ
ΔΗΜΟΣ ΝΟΤΙΑΣ ΚΥΝΟΥΡΙΑΣ
ΔΗΜΟΣ ΞΥΛΟΚΑΣΤΡΟΥ-ΕΥΡΩΣΤΙΝΗΣ
ΔΗΜΟΣ ΟΙΧΑΛΙΑΣ
ΔΗΜΟΣ ΠΥΛΟΥ-ΝΕΣΤΟΡΟΣ
ΔΗΜΟΣ ΣΙΚΥΩΝΙΩΝ
ΔΗΜΟΣ ΣΠΑΡΤΗΣ
ΔΗΜΟΣ ΤΡΙΠΟΛΗΣ
ΔΗΜΟΣ ΤΡΙΦΥΛΙΑΣ

5)

To retrieve, for each municipality of the Region of Peloponnese, its official name and all administrative divisions of Greece to which it belongs under the Kallikratis plan, we use property paths over the transitive “belongs to” relation (gag:ανήκει_σε). This allows us to avoid making any assumption about the number of administrative levels (e.g., municipal units, regional units, regions, decentralized administrations, country).

Municipalities are instances of `gag:Δήμος`, regions of `gag:Περιφέρεια`, and official names are given by the datatype property `gag:έχει_επίσημο_όνομα`.

The following SPARQL query provides the required result:

```
PREFIX gag: <http://geo.linkedopendata.gr/gag/ontology/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT DISTINCT ?municipalityName ?admName
WHERE {
  ?pelRegion rdf:type gag:Περιφέρεια ;
    gag:έχει_επίσημο_όνομα ?regionName .
  FILTER(CONTAINS(STR(?regionName), "ΠΕΛΟΠΟΝΝΗΣΟΥ"))

  ?municipality rdf:type gag:Δήμος ;
    gag:έχει_επίσημο_όνομα ?municipalityName .
  ?municipality gag:ανήκει_σε+ ?pelRegion .

  ?municipality gag:ανήκει_σε+ ?admUnit .
  ?admUnit gag:έχει_επίσημο_όνομα ?admName .
}
ORDER BY ?municipalityName ?admName
```

This query first identifies the Region of Peloponnese, then selects all municipalities that transitively belong to this region, and finally, using another transitive property path, retrieves every administrative unit in their containment hierarchy. The result is a list of municipality names, each paired with the names of all administrative divisions it belongs to, without relying on any explicit knowledge of the number or ordering of Kallikratis administration levels.

Column1	Column2
municipalityName	admName
ΔΗΜΟΣ ΑΝΑΤΟΛΙΚΗΣ ΜΑΝΗΣ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΑΝΑΤΟΛΙΚΗΣ ΜΑΝΗΣ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΑΝΑΤΟΛΙΚΗΣ ΜΑΝΗΣ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΑΝΑΤΟΛΙΚΗΣ ΜΑΝΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΑΚΩΝΙΑΣ
ΔΗΜΟΣ ΑΡΓΟΥΣ-ΜΥΚΗΝΩΝ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΑΡΓΟΥΣ-ΜΥΚΗΝΩΝ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΑΡΓΟΥΣ-ΜΥΚΗΝΩΝ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΑΡΓΟΥΣ-ΜΥΚΗΝΩΝ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΓΟΛΙΔΑΣ
ΔΗΜΟΣ ΒΕΛΟΥ-ΒΟΧΑΣ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΒΕΛΟΥ-ΒΟΧΑΣ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΒΕΛΟΥ-ΒΟΧΑΣ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΒΕΛΟΥ-ΒΟΧΑΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΟΡΙΝΘΙΑΣ
ΔΗΜΟΣ ΒΟΡΕΙΑΣ ΚΥΝΟΥΡΙΑΣ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΒΟΡΕΙΑΣ ΚΥΝΟΥΡΙΑΣ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΒΟΡΕΙΑΣ ΚΥΝΟΥΡΙΑΣ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΒΟΡΕΙΑΣ ΚΥΝΟΥΡΙΑΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΚΑΔΙΑΣ

ΔΗΜΟΣ ΓΟΡΤΥΝΙΑΣ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΓΟΡΤΥΝΙΑΣ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΓΟΡΤΥΝΙΑΣ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΓΟΡΤΥΝΙΑΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΚΑΔΙΑΣ
ΔΗΜΟΣ ΔΥΤΙΚΗΣ ΜΑΝΗΣ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΔΥΤΙΚΗΣ ΜΑΝΗΣ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΔΥΤΙΚΗΣ ΜΑΝΗΣ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΔΥΤΙΚΗΣ ΜΑΝΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΕΣΣΗΝΙΑΣ
ΔΗΜΟΣ ΕΛΑΦΟΝΗΣΟΥ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΕΛΑΦΟΝΗΣΟΥ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΕΛΑΦΟΝΗΣΟΥ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΕΛΑΦΟΝΗΣΟΥ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΑΚΩΝΙΑΣ
ΔΗΜΟΣ ΕΠΙΔΑΥΡΟΥ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΕΠΙΔΑΥΡΟΥ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΕΠΙΔΑΥΡΟΥ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΕΠΙΔΑΥΡΟΥ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΓΟΛΙΔΑΣ
ΔΗΜΟΣ ΕΡΜΙΟΝΙΔΑΣ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΕΡΜΙΟΝΙΔΑΣ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΕΡΜΙΟΝΙΔΑΣ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΕΡΜΙΟΝΙΔΑΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΓΟΛΙΔΑΣ
ΔΗΜΟΣ ΕΥΡΩΤΑ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΕΥΡΩΤΑ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΕΥΡΩΤΑ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΕΥΡΩΤΑ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΑΚΩΝΙΑΣ
ΔΗΜΟΣ ΚΑΛΑΜΑΤΑΣ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΚΑΛΑΜΑΤΑΣ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΚΑΛΑΜΑΤΑΣ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΚΑΛΑΜΑΤΑΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΕΣΣΗΝΙΑΣ
ΔΗΜΟΣ ΚΟΡΙΝΘΙΩΝ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΚΟΡΙΝΘΙΩΝ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΚΟΡΙΝΘΙΩΝ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΚΟΡΙΝΘΙΩΝ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΟΡΙΝΘΙΑΣ
ΔΗΜΟΣ ΛΟΥΤΡΑΚΙΟΥ-ΑΓ.ΘΕΟΔΩΡΩΝ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΛΟΥΤΡΑΚΙΟΥ-ΑΓ.ΘΕΟΔΩΡΩΝ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΛΟΥΤΡΑΚΙΟΥ-ΑΓ.ΘΕΟΔΩΡΩΝ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΛΟΥΤΡΑΚΙΟΥ-ΑΓ.ΘΕΟΔΩΡΩΝ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΟΡΙΝΘΙΑΣ
ΔΗΜΟΣ ΜΕΓΑΛΟΠΟΛΗΣ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΜΕΓΑΛΟΠΟΛΗΣ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΜΕΓΑΛΟΠΟΛΗΣ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΜΕΓΑΛΟΠΟΛΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΚΑΔΙΑΣ
ΔΗΜΟΣ ΜΕΣΣΗΝΗΣ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΜΕΣΣΗΝΗΣ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΜΕΣΣΗΝΗΣ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΜΕΣΣΗΝΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΕΣΣΗΝΙΑΣ

ΔΗΜΟΣ ΜΟΝΕΜΒΑΣΙΑΣ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΜΟΝΕΜΒΑΣΙΑΣ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΜΟΝΕΜΒΑΣΙΑΣ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΜΟΝΕΜΒΑΣΙΑΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΑΚΩΝΙΑΣ
ΔΗΜΟΣ ΝΑΥΠΛΙΕΩΝ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΝΑΥΠΛΙΕΩΝ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΝΑΥΠΛΙΕΩΝ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΝΑΥΠΛΙΕΩΝ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΓΟΛΙΔΑΣ
ΔΗΜΟΣ ΝΕΜΕΑΣ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΝΕΜΕΑΣ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΝΕΜΕΑΣ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΝΕΜΕΑΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΟΡΙΝΘΙΑΣ
ΔΗΜΟΣ ΝΟΤΙΑΣ ΚΥΝΟΥΡΙΑΣ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΝΟΤΙΑΣ ΚΥΝΟΥΡΙΑΣ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΝΟΤΙΑΣ ΚΥΝΟΥΡΙΑΣ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΝΟΤΙΑΣ ΚΥΝΟΥΡΙΑΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΚΑΔΙΑΣ
ΔΗΜΟΣ ΞΥΛΟΚΑΣΤΡΟΥ-ΕΥΡΩΣΤΙΝΗΣ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΞΥΛΟΚΑΣΤΡΟΥ-ΕΥΡΩΣΤΙΝΗΣ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΞΥΛΟΚΑΣΤΡΟΥ-ΕΥΡΩΣΤΙΝΗΣ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΞΥΛΟΚΑΣΤΡΟΥ-ΕΥΡΩΣΤΙΝΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΟΡΙΝΘΙΑΣ
ΔΗΜΟΣ ΟΙΧΑΛΙΑΣ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΟΙΧΑΛΙΑΣ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΟΙΧΑΛΙΑΣ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΟΙΧΑΛΙΑΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΕΣΣΗΝΙΑΣ
ΔΗΜΟΣ ΠΥΛΟΥ-ΝΕΣΤΟΡΟΣ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΠΥΛΟΥ-ΝΕΣΤΟΡΟΣ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΠΥΛΟΥ-ΝΕΣΤΟΡΟΣ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΠΥΛΟΥ-ΝΕΣΤΟΡΟΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΕΣΣΗΝΙΑΣ
ΔΗΜΟΣ ΣΙΚΥΩΝΙΩΝ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΣΙΚΥΩΝΙΩΝ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΣΙΚΥΩΝΙΩΝ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΣΙΚΥΩΝΙΩΝ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΟΡΙΝΘΙΑΣ
ΔΗΜΟΣ ΣΠΑΡΤΗΣ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΣΠΑΡΤΗΣ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΣΠΑΡΤΗΣ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΣΠΑΡΤΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΑΚΩΝΙΑΣ
ΔΗΜΟΣ ΤΡΙΠΟΛΗΣ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΤΡΙΠΟΛΗΣ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΤΡΙΠΟΛΗΣ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΤΡΙΠΟΛΗΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΚΑΔΙΑΣ
ΔΗΜΟΣ ΤΡΙΦΥΛΙΑΣ	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΩΝ
ΔΗΜΟΣ ΤΡΙΦΥΛΙΑΣ	ΕΛΛΑΔΑ
ΔΗΜΟΣ ΤΡΙΦΥΛΙΑΣ	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
ΔΗΜΟΣ ΤΡΙΦΥΛΙΑΣ	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΕΣΣΗΝΙΑΣ

6)

In the Kallikratis ontology, regions, regional units and municipalities are represented by the classes `gag:Περιφέρεια`, `gag:Περιφερειακή_Ενότητα`, and `gag:Δήμος`, respectively. Administrative containment is modeled by the object property `gag:ανήκει_σε`, and official names are provided by the datatype property `gag:έχει_επίσημο_όνομα`.

To obtain, for each region of Greece, (i) its official name, (ii) the number of municipalities that belong to it, (iii) the official name of each regional unit that belongs to it, and (iv) the number of municipalities that belong to that regional unit, we use SPARQL 1.1 aggregates and subqueries.

The following query first counts municipalities per region, then counts municipalities per regional unit within each region, and finally joins these counts with the human-readable names:

```
PREFIX gag: <http://geo.linkedopendata.gr/gag/ontology/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT ?regionName ?regionMunicipalities ?unitName ?unitMunicipalities
WHERE {

  # Region name
  ?region rdf:type gag:Περιφέρεια ;
    gag:έχει_επίσημο_όνομα ?regionName .

  # Subquery: number of municipalities per region
  {
    SELECT ?region (COUNT(DISTINCT ?mun) AS ?regionMunicipalities)
    WHERE {
      ?unit rdf:type gag:Περιφερειακή_Ενότητα ;
        gag:ανήκει_σε ?region .
      ?mun rdf:type gag:Δήμος ;
        gag:ανήκει_σε ?unit .
    }
    GROUP BY ?region
  }

  # Subquery: number of municipalities per regional unit (within each region)
  {
    SELECT ?region ?unitName (COUNT(DISTINCT ?mun2) AS ?unitMunicipalities)
    WHERE {
      ?unit rdf:type gag:Περιφερειακή_Ενότητα ;
        gag:ανήκει_σε ?region ;
        gag:έχει_επίσημο_όνομα ?unitName .
      ?mun2 rdf:type gag:Δήμος ;
        gag:ανήκει_σε ?unit .
    }
    GROUP BY ?region ?unitName
  }
}
ORDER BY ?regionName ?unitName
```

regionName	regionMunicipalities	unitName
ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ	22	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΔΡΑΜΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ	22	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΕΒΡΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ	22	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΘΑΣΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ	22	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΑΒΑΛΑΣ

ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ	22	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΞΑΝΘΗΣ
ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ	22	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΡΟΔΟΠΗΣ
ΠΕΡΙΦΕΡΕΙΑ ΑΤΤΙΚΗΣ	66	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΝΑΤΟΛΙΚΗΣ ΑΤΤΙΚΗΣ
ΠΕΡΙΦΕΡΕΙΑ ΑΤΤΙΚΗΣ	66	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΒΟΡΕΙΟΥ ΤΟΜΕΑ ΑΤΤΙΚΗΣ
ΠΕΡΙΦΕΡΕΙΑ ΑΤΤΙΚΗΣ	66	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΔΥΤΙΚΗΣ ΑΤΤΙΚΗΣ
ΠΕΡΙΦΕΡΕΙΑ ΑΤΤΙΚΗΣ	66	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΔΥΤΙΚΟΥ ΤΟΜΕΑ ΑΤΤΙΚΗΣ
ΠΕΡΙΦΕΡΕΙΑ ΑΤΤΙΚΗΣ	66	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΕΝΤΡΙΚΟΥ ΤΟΜΕΑ ΑΤΤΙΚΗΣ
ΠΕΡΙΦΕΡΕΙΑ ΑΤΤΙΚΗΣ	66	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΝΗΣΩΝ
ΠΕΡΙΦΕΡΕΙΑ ΑΤΤΙΚΗΣ	66	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΝΟΤΙΟΥ ΤΟΜΕΑ ΑΤΤΙΚΗΣ
ΠΕΡΙΦΕΡΕΙΑ ΑΤΤΙΚΗΣ	66	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΠΕΙΡΑΙΩΣ
ΠΕΡΙΦΕΡΕΙΑ ΒΟΡΕΙΟΥ ΑΙΓΑΙΟΥ	9	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΙΚΑΡΙΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΒΟΡΕΙΟΥ ΑΙΓΑΙΟΥ	9	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΕΣΒΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΒΟΡΕΙΟΥ ΑΙΓΑΙΟΥ	9	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΗΜΝΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΒΟΡΕΙΟΥ ΑΙΓΑΙΟΥ	9	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΣΑΜΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΒΟΡΕΙΟΥ ΑΙΓΑΙΟΥ	9	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΧΙΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ	19	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΙΤΩΛΟΑΚΑΡΝΑΝ
ΠΕΡΙΦΕΡΕΙΑ ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ	19	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΧΑΪΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ	19	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΗΛΕΙΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΔΥΤΙΚΗΣ ΜΑΚΕΔΟΝΙΑΣ	12	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΓΡΕΒΕΝΩΝ
ΠΕΡΙΦΕΡΕΙΑ ΔΥΤΙΚΗΣ ΜΑΚΕΔΟΝΙΑΣ	12	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΑΣΤΟΡΙΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΔΥΤΙΚΗΣ ΜΑΚΕΔΟΝΙΑΣ	12	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΟΖΑΝΗΣ
ΠΕΡΙΦΕΡΕΙΑ ΔΥΤΙΚΗΣ ΜΑΚΕΔΟΝΙΑΣ	12	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΦΛΩΡΙΝΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΗΠΕΙΡΟΥ	18	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΤΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΗΠΕΙΡΟΥ	18	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΘΕΣΠΡΩΤΙΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΗΠΕΙΡΟΥ	18	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΙΩΑΝΝΙΝΩΝ
ΠΕΡΙΦΕΡΕΙΑ ΗΠΕΙΡΟΥ	18	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΠΡΕΒΕΖΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΘΕΣΣΑΛΙΑΣ	25	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΑΡΔΙΤΣΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΘΕΣΣΑΛΙΑΣ	25	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΑΡΙΣΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΘΕΣΣΑΛΙΑΣ	25	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΑΓΝΗΣΙΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΘΕΣΣΑΛΙΑΣ	25	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΣΠΟΡΑΔΩΝ
ΠΕΡΙΦΕΡΕΙΑ ΘΕΣΣΑΛΙΑΣ	25	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΤΡΙΚΑΛΩΝ
ΠΕΡΙΦΕΡΕΙΑ ΙΟΝΙΩΝ ΝΗΣΩΝ	7	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΖΑΚΥΝΘΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΙΟΝΙΩΝ ΝΗΣΩΝ	7	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΙΘΑΚΗΣ
ΠΕΡΙΦΕΡΕΙΑ ΙΟΝΙΩΝ ΝΗΣΩΝ	7	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΕΡΚΥΡΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΙΟΝΙΩΝ ΝΗΣΩΝ	7	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΕΦΑΛΛΗΝΙΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΙΟΝΙΩΝ ΝΗΣΩΝ	7	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΕΥΚΑΔΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΚΕΝΤΡΙΚΗΣ ΜΑΚΕΔΟΝΙΑΣ	38	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΗΜΑΘΙΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΚΕΝΤΡΙΚΗΣ ΜΑΚΕΔΟΝΙΑΣ	38	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΘΕΣΣΑΛΟΝΙΚΗΣ
ΠΕΡΙΦΕΡΕΙΑ ΚΕΝΤΡΙΚΗΣ ΜΑΚΕΔΟΝΙΑΣ	38	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΙΛΚΙΣ
ΠΕΡΙΦΕΡΕΙΑ ΚΕΝΤΡΙΚΗΣ ΜΑΚΕΔΟΝΙΑΣ	38	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΠΕΛΛΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΚΕΝΤΡΙΚΗΣ ΜΑΚΕΔΟΝΙΑΣ	38	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΠΙΕΡΙΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΚΕΝΤΡΙΚΗΣ ΜΑΚΕΔΟΝΙΑΣ	38	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΣΕΡΡΩΝ
ΠΕΡΙΦΕΡΕΙΑ ΚΕΝΤΡΙΚΗΣ ΜΑΚΕΔΟΝΙΑΣ	38	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΧΑΛΚΙΔΙΚΗΣ
ΠΕΡΙΦΕΡΕΙΑ ΚΡΗΤΗΣ	24	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΗΡΑΚΛΕΙΟΥ

ΠΕΡΙΦΕΡΕΙΑ ΚΡΗΤΗΣ	24	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΑΣΙΘΙΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΚΡΗΤΗΣ	24	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΡΕΘΥΜΝΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΚΡΗΤΗΣ	24	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΧΑΝΙΩΝ
ΠΕΡΙΦΕΡΕΙΑ ΝΟΤΙΟΥ ΑΙΓΑΙΟΥ	34	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΝΔΡΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΝΟΤΙΟΥ ΑΙΓΑΙΟΥ	34	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΘΗΡΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΝΟΤΙΟΥ ΑΙΓΑΙΟΥ	34	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΑΛΥΜΝΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΝΟΤΙΟΥ ΑΙΓΑΙΟΥ	34	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΑΡΠΑΘΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΝΟΤΙΟΥ ΑΙΓΑΙΟΥ	34	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΕΑΣ-ΚΥΘΝΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΝΟΤΙΟΥ ΑΙΓΑΙΟΥ	34	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΩ
ΠΕΡΙΦΕΡΕΙΑ ΝΟΤΙΟΥ ΑΙΓΑΙΟΥ	34	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΗΛΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΝΟΤΙΟΥ ΑΙΓΑΙΟΥ	34	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΥΚΟΝΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΝΟΤΙΟΥ ΑΙΓΑΙΟΥ	34	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΝΑΞΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΝΟΤΙΟΥ ΑΙΓΑΙΟΥ	34	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΠΑΡΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΝΟΤΙΟΥ ΑΙΓΑΙΟΥ	34	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΡΟΔΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΝΟΤΙΟΥ ΑΙΓΑΙΟΥ	34	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΣΥΡΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΝΟΤΙΟΥ ΑΙΓΑΙΟΥ	34	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΤΗΝΟΥ
ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ	26	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΓΟΛΙΔΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ	26	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΚΑΔΙΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ	26	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΟΡΙΝΘΙΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ	26	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΑΚΩΝΙΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ	26	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΕΣΣΗΝΙΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΣΤΕΡΕΑΣ ΕΛΛΑΔΑΣ	25	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΒΟΙΩΤΙΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΣΤΕΡΕΑΣ ΕΛΛΑΔΑΣ	25	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΕΥΒΟΙΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΣΤΕΡΕΑΣ ΕΛΛΑΔΑΣ	25	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΕΥΡΥΤΑΝΙΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΣΤΕΡΕΑΣ ΕΛΛΑΔΑΣ	25	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΦΘΙΩΤΙΔΑΣ
ΠΕΡΙΦΕΡΕΙΑ ΣΤΕΡΕΑΣ ΕΛΛΑΔΑΣ	25	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΦΩΚΙΔΑΣ

7)

In the Kallikratis ontology, administrative containment between units is represented by the object property `gag:ανήκει_σε`, while populations are given by the datatype property `gag:έχει_πληθυσμό`. To check the consistency of the dataset with respect to stated populations, we must verify that, for every administrative unit *B* at some level *L+1*, the sum of the populations of all administrative units *A* at level *L* that belong to *B* equals the population of *B* itself.

This can be done with a single SPARQL 1.1 query that is independent of the specific levels of the Kallikratis hierarchy.

The query groups all child units by their parent, sums their populations, and compares this sum with the population stated for the parent.

Any parent for which these two values differ represents an inconsistency.

```

PREFIX gag: <http://geo.linkedopendata.gr/gag/ontology/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

SELECT ?parent ?parentName ?parentPop
      (SUM(xsd:integer(?childPop)) AS ?childrenPop)
WHERE {
  # Parent administrative unit with stated population (level L+1)

```

```

?parent gag:έχει_πληθυσμό ?parentPop .
OPTIONAL { ?parent gag:έχει_επίσημο_όνομα ?parentName . }

# Child administrative units of level L that belong to this parent
?child gag:ανήκει_σε ?parent ;
    gag:έχει_πληθυσμό ?childPop .
}
GROUP BY ?parent ?parentName ?parentPop
HAVING ( xsd:integer(?parentPop) != SUM(xsd:integer(?childPop)) )
ORDER BY ?parentName

```

If this query returns no results, the population figures in the dataset are fully consistent with the hierarchical structure.

parent	parentName
http://geo.linkedopendata.gr/gag/id/996	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΑΙΓΑΙΟΥ
http://geo.linkedopendata.gr/gag/id/991	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΑΤΤΙΚΗΣ
http://geo.linkedopendata.gr/gag/id/993	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΗΠΕΙΡΟΥ-ΔΥΤΙΚΗΣ ΜΑΚΕΔΟΝΙΑΣ
http://geo.linkedopendata.gr/gag/id/994	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΘΕΣΣΑΛΙΑΣ-ΣΤΕΡΕΑΣ ΕΛΛΑΔΑΣ
http://geo.linkedopendata.gr/gag/id/997	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΚΡΗΤΗΣ
http://geo.linkedopendata.gr/gag/id/992	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΜΑΚΕΔΟΝΙΑΣ-ΘΡΑΚΗΣ
http://geo.linkedopendata.gr/gag/id/995	ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ
http://geo.linkedopendata.gr/gag/id/0	ΕΛΛΑΔΑ
http://geo.linkedopendata.gr/gag/id/1	ΠΕΡΙΦΕΡΕΙΑ ΑΝ. ΜΑΚΕΔΟΝΙΑΣ ΘΡΑΚΗΣ
http://geo.linkedopendata.gr/gag/id/9	ΠΕΡΙΦΕΡΕΙΑ ΑΤΤΙΚΗΣ
http://geo.linkedopendata.gr/gag/id/11	ΠΕΡΙΦΕΡΕΙΑ ΒΟΡΕΙΟΥ ΑΙΓΑΙΟΥ
http://geo.linkedopendata.gr/gag/id/7	ΠΕΡΙΦΕΡΕΙΑ ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ
http://geo.linkedopendata.gr/gag/id/3	ΠΕΡΙΦΕΡΕΙΑ ΔΥΤΙΚΗΣ ΜΑΚΕΔΟΝΙΑΣ
http://geo.linkedopendata.gr/gag/id/4	ΠΕΡΙΦΕΡΕΙΑ ΗΠΕΙΡΟΥ
http://geo.linkedopendata.gr/gag/id/5	ΠΕΡΙΦΕΡΕΙΑ ΘΕΣΣΑΛΙΑΣ
http://geo.linkedopendata.gr/gag/id/6	ΠΕΡΙΦΕΡΕΙΑ ΙΟΝΙΩΝ ΝΗΣΩΝ
http://geo.linkedopendata.gr/gag/id/2	ΠΕΡΙΦΕΡΕΙΑ ΚΕΝΤΡΙΚΗΣ ΜΑΚΕΔΟΝΙΑΣ
http://geo.linkedopendata.gr/gag/id/13	ΠΕΡΙΦΕΡΕΙΑ ΚΡΗΤΗΣ
http://geo.linkedopendata.gr/gag/id/12	ΠΕΡΙΦΕΡΕΙΑ ΝΟΤΙΟΥ ΑΙΓΑΙΟΥ
http://geo.linkedopendata.gr/gag/id/10	ΠΕΡΙΦΕΡΕΙΑ ΠΕΛΟΠΟΝΝΗΣΟΥ
http://geo.linkedopendata.gr/gag/id/8	ΠΕΡΙΦΕΡΕΙΑ ΣΤΕΡΕΑΣ ΕΛΛΑΔΑΣ
http://geo.linkedopendata.gr/gag/id/701	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΙΤΩΛΟΑΚΑΡΝΑΝΙΑΣ
http://geo.linkedopendata.gr/gag/id/907	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΝΑΤΟΛΙΚΗΣ ΑΤΤΙΚΗΣ
http://geo.linkedopendata.gr/gag/id/1201	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΝΔΡΟΥ
http://geo.linkedopendata.gr/gag/id/1001	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΓΟΛΙΔΑΣ
http://geo.linkedopendata.gr/gag/id/1002	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΚΑΔΙΑΣ
http://geo.linkedopendata.gr/gag/id/401	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΡΤΑΣ
http://geo.linkedopendata.gr/gag/id/702	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΑΧΑΪΑΣ
http://geo.linkedopendata.gr/gag/id/801	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΒΟΙΩΤΙΑΣ
http://geo.linkedopendata.gr/gag/id/901	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΒΟΡΕΙΟΥ ΤΟΜΕΑ ΑΘΗΝΩΝ
http://geo.linkedopendata.gr/gag/id/301	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΓΡΕΒΕΝΩΝ
http://geo.linkedopendata.gr/gag/id/101	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΔΡΑΜΑΣ
http://geo.linkedopendata.gr/gag/id/908	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΔΥΤΙΚΗΣ ΑΤΤΙΚΗΣ

http://geo.linkedopendata.gr/gag/id/902	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΔΥΤΙΚΟΥ ΤΟΜΕΑ ΑΘΗΝΩΝ
http://geo.linkedopendata.gr/gag/id/102	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΕΒΡΟΥ
http://geo.linkedopendata.gr/gag/id/802	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΕΥΒΟΙΑΣ
http://geo.linkedopendata.gr/gag/id/803	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΕΥΡΥΤΑΝΙΑΣ
http://geo.linkedopendata.gr/gag/id/601	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΖΑΚΥΝΘΟΥ
http://geo.linkedopendata.gr/gag/id/703	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΗΛΕΙΑΣ
http://geo.linkedopendata.gr/gag/id/201	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΗΜΑΘΙΑΣ
http://geo.linkedopendata.gr/gag/id/1301	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΗΡΑΚΛΕΙΟΥ
http://geo.linkedopendata.gr/gag/id/103	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΘΑΣΟΥ
http://geo.linkedopendata.gr/gag/id/402	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΘΕΣΠΡΩΤΙΑΣ
http://geo.linkedopendata.gr/gag/id/202	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΘΕΣΣΑΛΟΝΙΚΗΣ
http://geo.linkedopendata.gr/gag/id/1202	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΘΗΡΑΣ
http://geo.linkedopendata.gr/gag/id/602	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΙΘΑΚΗΣ
http://geo.linkedopendata.gr/gag/id/1101	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΙΚΑΡΙΑΣ
http://geo.linkedopendata.gr/gag/id/403	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΙΩΑΝΝΙΝΩΝ
http://geo.linkedopendata.gr/gag/id/104	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΑΒΑΛΑΣ
http://geo.linkedopendata.gr/gag/id/1203	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΑΛΥΜΝΟΥ
http://geo.linkedopendata.gr/gag/id/501	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΑΡΔΙΤΣΑΣ
http://geo.linkedopendata.gr/gag/id/1204	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΑΡΠΑΘΟΥ
http://geo.linkedopendata.gr/gag/id/302	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΑΣΤΟΡΙΑΣ
http://geo.linkedopendata.gr/gag/id/1205	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΕΑΣ-ΚΥΘΝΟΥ
http://geo.linkedopendata.gr/gag/id/903	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΕΝΤΡΙΚΟΥ ΤΟΜΕΑ ΑΘΗΝΩΝ
http://geo.linkedopendata.gr/gag/id/603	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΕΡΚΥΡΑΣ
http://geo.linkedopendata.gr/gag/id/604	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΕΦΑΛΛΗΝΙΑΣ
http://geo.linkedopendata.gr/gag/id/203	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΙΛΚΙΣ
http://geo.linkedopendata.gr/gag/id/303	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΟΖΑΝΗΣ
http://geo.linkedopendata.gr/gag/id/1003	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΟΡΙΝΘΙΑΣ
http://geo.linkedopendata.gr/gag/id/1206	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΚΩ
http://geo.linkedopendata.gr/gag/id/1004	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΑΚΩΝΙΑΣ
http://geo.linkedopendata.gr/gag/id/502	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΑΡΙΣΑΣ
http://geo.linkedopendata.gr/gag/id/1302	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΑΣΙΘΙΟΥ
http://geo.linkedopendata.gr/gag/id/1102	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΕΣΒΟΥ
http://geo.linkedopendata.gr/gag/id/605	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΕΥΚΑΔΑΣ
http://geo.linkedopendata.gr/gag/id/1103	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΛΗΜΝΟΥ
http://geo.linkedopendata.gr/gag/id/503	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΑΓΝΗΣΙΑΣ
http://geo.linkedopendata.gr/gag/id/1005	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΕΣΣΗΝΙΑΣ
http://geo.linkedopendata.gr/gag/id/1207	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΗΛΟΥ
http://geo.linkedopendata.gr/gag/id/1208	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΜΥΚΟΝΟΥ
http://geo.linkedopendata.gr/gag/id/1209	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΝΑΞΟΥ
http://geo.linkedopendata.gr/gag/id/906	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΝΗΣΩΝ
http://geo.linkedopendata.gr/gag/id/904	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΝΟΤΙΟΥ ΤΟΜΕΑ ΑΘΗΝΩΝ
http://geo.linkedopendata.gr/gag/id/105	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΞΑΝΘΗΣ
http://geo.linkedopendata.gr/gag/id/1210	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΠΑΡΟΥ
http://geo.linkedopendata.gr/gag/id/905	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΠΕΙΡΑΙΩΣ

http://geo.linkedopendata.gr/gag/id/204	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΠΕΛΛΑΣ
http://geo.linkedopendata.gr/gag/id/205	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΠΙΕΡΙΑΣ
http://geo.linkedopendata.gr/gag/id/404	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΠΡΕΒΕΖΑΣ
http://geo.linkedopendata.gr/gag/id/1303	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΡΕΘΥΜΝΟΥ
http://geo.linkedopendata.gr/gag/id/106	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΡΟΔΟΠΗΣ
http://geo.linkedopendata.gr/gag/id/1211	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΡΟΔΟΥ
http://geo.linkedopendata.gr/gag/id/1104	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΣΑΜΟΥ
http://geo.linkedopendata.gr/gag/id/206	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΣΕΡΡΩΝ
http://geo.linkedopendata.gr/gag/id/504	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΣΠΟΡΑΔΩΝ
http://geo.linkedopendata.gr/gag/id/1212	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΣΥΡΟΥ
http://geo.linkedopendata.gr/gag/id/1213	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΤΗΝΟΥ
http://geo.linkedopendata.gr/gag/id/505	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΤΡΙΚΑΛΩΝ
http://geo.linkedopendata.gr/gag/id/804	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΦΘΙΩΤΙΔΑΣ
http://geo.linkedopendata.gr/gag/id/304	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΦΛΩΡΙΝΑΣ
http://geo.linkedopendata.gr/gag/id/805	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΦΩΚΙΔΑΣ
http://geo.linkedopendata.gr/gag/id/207	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΧΑΛΚΙΔΙΚΗΣ
http://geo.linkedopendata.gr/gag/id/1304	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΧΑΝΙΩΝ
http://geo.linkedopendata.gr/gag/id/1105	ΠΕΡΙΦΕΡΕΙΑΚΗ ΕΝΟΤΗΤΑ ΧΙΟΥ

The query reveals that the dataset is not population-consistent, since the stated population of multiple administrative units does not match the sum of the populations of their subordinate units.

8)

In the Kallikratis dataset, decentralized administrations are represented by the class `gag:Αποκεντρωμένη_Διοίκηση`, and regional units by the class `gag:Περιφερειακή_Ενότητα`. Administrative containment is encoded using the property `gag:ανήκει_σε`, while official names are given by `gag:έχει_επίσημο_όνομα`.

Since the query must be expressed *without* SPARQL 1.1 aggregate operators, we identify decentralized administrations that consist of more than two regional units by checking whether at least three distinct regional units belong to the same decentralized administration.

This is done by introducing three variables (`?unit1`, `?unit2`, `?unit3`) and enforcing pairwise inequality.

The SPARQL query is:

```
PREFIX gag: <http://geo.linkedopendata.gr/gag/ontology/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT DISTINCT ?daName
WHERE {
  ?da rdf:type gag:Αποκεντρωμένη_Διοίκηση ;
    gag:έχει_επίσημο_όνομα ?daName .

  ?unit1 rdf:type gag:Περιφερειακή_Ενότητα ;
    gag:ανήκει_σε ?da .
  ?unit2 rdf:type gag:Περιφερειακή_Ενότητα ;
    gag:ανήκει_σε ?da .
  ?unit3 rdf:type gag:Περιφερειακή_Ενότητα ;
```

```
gag:ανήκει_σε ?da .

FILTER(?unit1 != ?unit2)
FILTER(?unit1 != ?unit3)
FILTER(?unit2 != ?unit3)
}
ORDER BY ?daName
```

This query returns exactly the decentralized administrations of Greece that consist of more than two regional units, without relying on any aggregate operators.

daName
"ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΑΙΓΑΙΟΥ"
"ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΑΤΤΙΚΗΣ"
"ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΗΠΕΙΡΟΥ-ΔΥΤΙΚΗΣ ΜΑΚΕΔΟΝΙΑΣ"
"ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΘΕΣΣΑΛΙΑΣ-ΣΤΕΡΕΑΣ ΕΛΛΑΔΑΣ"
"ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΚΡΗΤΗΣ"
"ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΜΑΚΕΔΟΝΙΑΣ-ΘΡΑΚΗΣ"
"ΑΠΟΚΕΝΤΡΩΜΕΝΗ ΔΙΟΙΚΗΣΗ ΠΕΛΟΠΟΝΝΗΣΟΥ-ΔΥΤΙΚΗΣ ΕΛΛΑΔΑΣ-ΙΟΝΙΟΥ"

Exercise 3

1)

Using the current version of the schema.org core ontology loaded in GraphDB with RDFS reasoning enabled, the subclasses of schema:CollegeOrUniversity were retrieved with the following SPARQL 1.1 query:

```
PREFIX schema: <https://schema.org/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT DISTINCT ?subclass
WHERE {
  ?subclass rdfs:subClassOf+ schema:CollegeOrUniversity .
}
```

The query returns only:

subclass
schema:CollegeOrUniversity

Therefore, in this version of the schema.org ontology there are **no declared proper subclasses** of the class (type) CollegeOrUniversity; the only subclass entailed is the class itself.

2)

To compute the **superclasses** of CollegeOrUniversity in the schema.org ontology, we exploit the fact that rdfs:subClassOf is transitive. Using SPARQL 1.1 property paths, we ask for all classes that schema:CollegeOrUniversity is (directly or indirectly) a subclass of.

```
PREFIX schema: <https://schema.org/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
```

```

SELECT DISTINCT ?superclass
WHERE {
  schema:CollegeOrUniversity rdfs:subClassOf+ ?superclass .
}

```

superclass
schema:Organization
schema:Thing
schema:CollegeOrUniversity
schema:Place
schema:EducationalOrganization
schema:CivicStructure

3)

All properties defined for schema:CollegeOrUniversity and inherited from its superclasses are exactly those properties ?property for which there exists a class ?domain such that:

schema:CollegeOrUniversity rdfs:subClassOf* ?domain (i.e., ?domain is CollegeOrUniversity itself or one of its superclasses), and

?property schema:domainIncludes ?domain.

These properties are retrieved by the following SPARQL 1.1 query:

```

PREFIX schema: <https://schema.org/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT DISTINCT ?property ?domain
WHERE {
  # CollegeOrUniversity and all its superclasses
  schema:CollegeOrUniversity rdfs:subClassOf* ?domain .

  # Properties whose (schema.org) domain includes one of these classes
  ?property schema:domainIncludes ?domain .
}
ORDER BY ?domain ?property

```

When evaluated over the schema.org core ontology (with RDFS reasoning enabled), this query returns precisely the set of properties defined for CollegeOrUniversity and all properties inherited from its superclasses.

Column1	Column2
property	domain
https://schema.org/openingHours	https://schema.org/CivicStructure
https://schema.org/alumni	https://schema.org/EducationalOrganization
https://schema.org/acceptedPaymentMethod	https://schema.org/Organization
https://schema.org/actionableFeedbackPolicy	https://schema.org/Organization
https://schema.org/address	https://schema.org/Organization
https://schema.org/agentInteractionStatistic	https://schema.org/Organization
https://schema.org/aggregateRating	https://schema.org/Organization

https://schema.org/alumni	https://schema.org/Organization
https://schema.org/areaServed	https://schema.org/Organization
https://schema.org/award	https://schema.org/Organization
https://schema.org/awards	https://schema.org/Organization
https://schema.org/brand	https://schema.org/Organization
https://schema.org/companyRegistration	https://schema.org/Organization
https://schema.org/contactPoint	https://schema.org/Organization
https://schema.org/contactPoints	https://schema.org/Organization
https://schema.org/correctionsPolicy	https://schema.org/Organization
https://schema.org/department	https://schema.org/Organization
https://schema.org/dissolutionDate	https://schema.org/Organization
https://schema.org/diversityPolicy	https://schema.org/Organization
https://schema.org/diversityStaffingReport	https://schema.org/Organization
https://schema.org/duns	https://schema.org/Organization
https://schema.org/email	https://schema.org/Organization
https://schema.org/employee	https://schema.org/Organization
https://schema.org/employees	https://schema.org/Organization
https://schema.org/ethicsPolicy	https://schema.org/Organization
https://schema.org/event	https://schema.org/Organization
https://schema.org/events	https://schema.org/Organization
https://schema.org/faxNumber	https://schema.org/Organization
https://schema.org/founder	https://schema.org/Organization
https://schema.org/founders	https://schema.org/Organization
https://schema.org/foundingDate	https://schema.org/Organization
https://schema.org/foundingLocation	https://schema.org/Organization
https://schema.org/funder	https://schema.org/Organization
https://schema.org/funding	https://schema.org/Organization
https://schema.org/globalLocationNumber	https://schema.org/Organization
https://schema.org/hasCertification	https://schema.org/Organization
https://schema.org/hasCredential	https://schema.org/Organization
https://schema.org/hasGS1DigitalLink	https://schema.org/Organization
https://schema.org/hasMemberProgram	https://schema.org/Organization
https://schema.org/hasMerchantReturnPolicy	https://schema.org/Organization
https://schema.org/hasOfferCatalog	https://schema.org/Organization
https://schema.org/hasPOS	https://schema.org/Organization
https://schema.org/hasProductReturnPolicy	https://schema.org/Organization
https://schema.org/hasShippingService	https://schema.org/Organization
https://schema.org/interactionStatistic	https://schema.org/Organization
https://schema.org/isicV4	https://schema.org/Organization
https://schema.org/iso6523Code	https://schema.org/Organization
https://schema.org/keywords	https://schema.org/Organization
https://schema.org/knownAbout	https://schema.org/Organization
https://schema.org/knownLanguage	https://schema.org/Organization
https://schema.org/legalAddress	https://schema.org/Organization

https://schema.org/legalName	https://schema.org/Organization
https://schema.org/legalRepresentative	https://schema.org/Organization
https://schema.org/leiCode	https://schema.org/Organization
https://schema.org/location	https://schema.org/Organization
https://schema.org/logo	https://schema.org/Organization
https://schema.org/makesOffer	https://schema.org/Organization
https://schema.org/member	https://schema.org/Organization
https://schema.org/memberOf	https://schema.org/Organization
https://schema.org/members	https://schema.org/Organization
https://schema.org/naics	https://schema.org/Organization
https://schema.org/nonprofitStatus	https://schema.org/Organization
https://schema.org/numberOfEmployees	https://schema.org/Organization
https://schema.org/ownershipFundingInfo	https://schema.org/Organization
https://schema.org/owns	https://schema.org/Organization
https://schema.org/parentOrganization	https://schema.org/Organization
https://schema.org/publishingPrinciples	https://schema.org/Organization
https://schema.org/review	https://schema.org/Organization
https://schema.org/reviews	https://schema.org/Organization
https://schema.org/seeks	https://schema.org/Organization
https://schema.org/serviceArea	https://schema.org/Organization
https://schema.org/skills	https://schema.org/Organization
https://schema.org/slogan	https://schema.org/Organization
https://schema.org/sponsor	https://schema.org/Organization
https://schema.org/subOrganization	https://schema.org/Organization
https://schema.org/taxID	https://schema.org/Organization
https://schema.org/telephone	https://schema.org/Organization
https://schema.org/unnamedSourcesPolicy	https://schema.org/Organization
https://schema.org/vatID	https://schema.org/Organization
https://schema.org/additionalProperty	https://schema.org/Place
https://schema.org/address	https://schema.org/Place
https://schema.org/aggregateRating	https://schema.org/Place
https://schema.org/amenityFeature	https://schema.org/Place
https://schema.org/branchCode	https://schema.org/Place
https://schema.org/containedIn	https://schema.org/Place
https://schema.org/containedInPlace	https://schema.org/Place
https://schema.org/containsPlace	https://schema.org/Place
https://schema.org/event	https://schema.org/Place
https://schema.org/events	https://schema.org/Place
https://schema.org/faxNumber	https://schema.org/Place
https://schema.org/geo	https://schema.org/Place
https://schema.org/geoContains	https://schema.org/Place
https://schema.org/geoCoveredBy	https://schema.org/Place
https://schema.org/geoCovers	https://schema.org/Place
https://schema.org/geoCrosses	https://schema.org/Place

https://schema.org/geoDisjoint	https://schema.org/Place
https://schema.org/geoEquals	https://schema.org/Place
https://schema.org/geoIntersects	https://schema.org/Place
https://schema.org/geoOverlaps	https://schema.org/Place
https://schema.org/geoTouches	https://schema.org/Place
https://schema.org/geoWithin	https://schema.org/Place
https://schema.org/globalLocationNumber	https://schema.org/Place
https://schema.org/hasCertification	https://schema.org/Place
https://schema.org/hasDriveThroughService	https://schema.org/Place
https://schema.org/hasGS1DigitalLink	https://schema.org/Place
https://schema.org/hasMap	https://schema.org/Place
https://schema.org/isAccessibleForFree	https://schema.org/Place
https://schema.org/isicV4	https://schema.org/Place
https://schema.org/keywords	https://schema.org/Place
https://schema.org/latitude	https://schema.org/Place
https://schema.org/logo	https://schema.org/Place
https://schema.org/longitude	https://schema.org/Place
https://schema.org/map	https://schema.org/Place
https://schema.org/maps	https://schema.org/Place
https://schema.org/maximumAttendeeCapacity	https://schema.org/Place
https://schema.org/openingHoursSpecification	https://schema.org/Place
https://schema.org/photo	https://schema.org/Place
https://schema.org/photos	https://schema.org/Place
https://schema.org/publicAccess	https://schema.org/Place
https://schema.org/review	https://schema.org/Place
https://schema.org/reviews	https://schema.org/Place
https://schema.org/slogan	https://schema.org/Place
https://schema.org/smokingAllowed	https://schema.org/Place
https://schema.org/specialOpeningHoursSpecification	https://schema.org/Place
https://schema.org/telephone	https://schema.org/Place
https://schema.org/tourBookingPage	https://schema.org/Place
https://schema.org/additionalType	https://schema.org/Thing
https://schema.org/alternateName	https://schema.org/Thing
https://schema.org/description	https://schema.org/Thing
https://schema.org/disambiguatingDescription	https://schema.org/Thing
https://schema.org/identifier	https://schema.org/Thing
https://schema.org/image	https://schema.org/Thing
https://schema.org/mainEntityOfPage	https://schema.org/Thing
https://schema.org/name	https://schema.org/Thing
https://schema.org/potentialAction	https://schema.org/Thing
https://schema.org/sameAs	https://schema.org/Thing
https://schema.org/subjectOf	https://schema.org/Thing
https://schema.org/url	https://schema.org/Thing

4)

All classes that are subclasses of schema:Thing and lie at most two subclass links away in the rdfs:subClassOf hierarchy are obtained by the following SPARQL 1.1 query:

```
PREFIX schema: <https://schema.org/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT DISTINCT ?class
WHERE {
  {
    # subclasses at distance 1 from Thing
    ?class rdfs:subClassOf schema:Thing .
  }
  UNION
  {
    # subclasses at distance 2 from Thing
    ?class rdfs:subClassOf ?mid .
    ?mid rdfs:subClassOf schema:Thing .
  }
  FILTER (?class != schema:Thing)
}
ORDER BY ?class
```

Evaluated over the schema.org core ontology in GraphDB (with RDFS reasoning enabled), this query returns precisely the classes that are subclasses of schema:Thing at depth 1 or 2 in the subclass hierarchy.

Column1	Column2
class	
http	//purl.bioontology.org/ontology/SNOMEDCT/105590001
http	//purl.bioontology.org/ontology/SNOMEDCT/116154003
http	//purl.bioontology.org/ontology/SNOMEDCT/277132007
http	//purl.bioontology.org/ontology/SNOMEDCT/387713003
http	//purl.bioontology.org/ontology/SNOMEDCT/410942007
http	//purl.bioontology.org/ontology/SNOMEDCT/50731006
http	//purl.bioontology.org/ontology/SNOMEDCT/51114001
http	//purl.bioontology.org/ontology/SNOMEDCT/63653004
http	//purl.org/dc/dcmitype/Dataset
http	//purl.org/dc/dcmitype/Event
http	//purl.org/dc/dcmitype/Image
http	//purl.org/dc/dcmitype/Text
http	//purl.org/ontology/bibo/Issue
http	//purl.org/ontology/bibo/Periodical
http	//rdfs.org/ns/void#Dataset
http	//www.w3.org/1999/02/22-rdf-syntax-ns#Property
http	//www.w3.org/2000/01/rdf-schema#Class
http	//www.w3.org/2000/01/rdf-schema#ContainerMembershipProperty
http	//www.w3.org/2000/01/rdf-schema#Datatype
http	//www.w3.org/ns/dcat#Catalog

http	//www.w3.org/ns/dcat#Dataset
http	//www.w3.org/ns/dcat#Distribution
http	//xmlns.com/foaf/0.1/Person
https	//schema.org/3DModel
https	//schema.org/AMRadioChannel
https	//schema.org/APIReference
https	//schema.org/AboutPage
https	//schema.org/AcceptAction
https	//schema.org/Accommodation
https	//schema.org/AccountingService
https	//schema.org/AchieveAction
https	//schema.org/Action
https	//schema.org/ActionAccessSpecification
https	//schema.org/ActionStatusType
https	//schema.org/ActivateAction
https	//schema.org/AddAction
https	//schema.org/AdministrativeArea
https	//schema.org/AdultEntertainment
https	//schema.org/AdultOrientedEnumeration
https	//schema.org/AdvertiserContentArticle
https	//schema.org/AggregateOffer
https	//schema.org/AggregateRating
https	//schema.org/AgreeAction
https	//schema.org/Airline
https	//schema.org/Airport
https	//schema.org/AlignmentObject
https	//schema.org/AllocateAction
https	//schema.org/AmpStory
https	//schema.org/AmusementPark
https	//schema.org/AnalysisNewsArticle
https	//schema.org/AnatomicalStructure
https	//schema.org/AnatomicalSystem
https	//schema.org/AnimalShelter
https	//schema.org/Answer
https	//schema.org/Apartment
https	//schema.org/ApartmentComplex
https	//schema.org/AppendAction
https	//schema.org/ApplyAction
https	//schema.org/ApprovedIndication
https	//schema.org/Aquarium
https	//schema.org/ArchiveComponent
https	//schema.org/ArchiveOrganization
https	//schema.org/ArriveAction
https	//schema.org/ArtGallery

https	//schema.org/Artery
https	//schema.org/Article
https	//schema.org/AskAction
https	//schema.org/AskPublicNewsArticle
https	//schema.org/AssessAction
https	//schema.org/AssignAction
https	//schema.org/Atlas
https	//schema.org/Attorney
https	//schema.org/Audience
https	//schema.org/AudioObject
https	//schema.org/AudioObjectSnapshot
https	//schema.org/Audiobook
https	//schema.org/AuthorizeAction
https	//schema.org/AutoBodyShop
https	//schema.org/AutoDealer
https	//schema.org/AutoPartsStore
https	//schema.org/AutoRental
https	//schema.org/AutoRepair
https	//schema.org/AutoWash
https	//schema.org/AutomatedTeller
https	//schema.org/AutomotiveBusiness
https	//schema.org/BackgroundNewsArticle
https	//schema.org/Bakery
https	//schema.org/BankAccount
https	//schema.org/BankOrCreditUnion
https	//schema.org/BarOrPub
https	//schema.org/Barcode
https	//schema.org/Beach
https	//schema.org/BeautySalon
https	//schema.org/BedAndBreakfast
https	//schema.org/BedDetails
https	//schema.org/BedType
https	//schema.org/BefriendAction
https	//schema.org/BikeStore
https	//schema.org/BioChemEntity
https	//schema.org/Blog
https	//schema.org/BlogPosting
https	//schema.org/BloodTest
https	//schema.org/BoardingPolicyType
https	//schema.org/BoatReservation
https	//schema.org/BoatTerminal
https	//schema.org/BoatTrip
https	//schema.org/BodyMeasurementTypeEnumeration
https	//schema.org/BodyOfWater

https	//schema.org/Bone
https	//schema.org/Book
https	//schema.org/BookFormatType
https	//schema.org/BookSeries
https	//schema.org/BookStore
https	//schema.org/BookmarkAction
https	//schema.org/BorrowAction
https	//schema.org/BowlingAlley
https	//schema.org/BrainStructure
https	//schema.org/Brand
https	//schema.org/BreadcrumbList
https	//schema.org/Brewery
https	//schema.org/Bridge
https	//schema.org/BroadcastChannel
https	//schema.org/BroadcastEvent
https	//schema.org/BroadcastFrequencySpecification
https	//schema.org/BroadcastService
https	//schema.org/BrokerageAccount
https	//schema.org/BuddhistTemple
https	//schema.org/BusOrCoach
https	//schema.org/BusReservation
https	//schema.org/BusStation
https	//schema.org/BusStop
https	//schema.org/BusTrip
https	//schema.org/BusinessAudience
https	//schema.org/BusinessEntityType
https	//schema.org/BusinessEvent
https	//schema.org/BusinessFunction
https	//schema.org/BuyAction
https	//schema.org/CDCPMDRecord
https	//schema.org/CableOrSatelliteService
https	//schema.org/CafeOrCoffeeShop
https	//schema.org/Campground
https	//schema.org/CampingPitch
https	//schema.org/Canal
https	//schema.org/CancelAction
https	//schema.org/Car
https	//schema.org/CarUsageType
https	//schema.org/Casino
https	//schema.org/CategoryCode
https	//schema.org/CategoryCodeSet
https	//schema.org/CatholicChurch
https	//schema.org/Cemetery
https	//schema.org/Certification

https	//schema.org/CertificationStatusEnumeration
https	//schema.org/Chapter
https	//schema.org/CheckAction
https	//schema.org/CheckInAction
https	//schema.org/CheckOutAction
https	//schema.org/CheckoutPage
https	//schema.org/ChemicalSubstance
https	//schema.org/ChildCare
https	//schema.org/ChildrensEvent
https	//schema.org/ChooseAction
https	//schema.org/Church
https	//schema.org/City
https	//schema.org/CityHall
https	//schema.org/CivicStructure
https	//schema.org/Claim
https	//schema.org/ClaimReview
https	//schema.org/Class
https	//schema.org/Clip
https	//schema.org/ClothingStore
https	//schema.org/Code
https	//schema.org/Collection
https	//schema.org/CollectionPage
https	//schema.org/CollegeOrUniversity
https	//schema.org/ComedyClub
https	//schema.org/ComedyEvent
https	//schema.org/ComicCoverArt
https	//schema.org/ComicIssue
https	//schema.org/ComicSeries
https	//schema.org/ComicStory
https	//schema.org/Comment
https	//schema.org/CommentAction
https	//schema.org/CommunicateAction
https	//schema.org/CommunityHealth
https	//schema.org/CompleteDataFeed
https	//schema.org/CompoundPriceSpecification
https	//schema.org/ComputerLanguage
https	//schema.org/ComputerStore
https	//schema.org/ConfirmAction
https	//schema.org/Consortium
https	//schema.org/ConstraintNode
https	//schema.org/ConsumeAction
https	//schema.org/ContactPage
https	//schema.org/ContactPoint
https	//schema.org/ContactPointOption

https	//schema.org/Continent
https	//schema.org/ControlAction
https	//schema.org/ConvenienceStore
https	//schema.org/Conversation
https	//schema.org/CookAction
https	//schema.org/Cooperative
https	//schema.org/Corporation
https	//schema.org/CorrectionComment
https	//schema.org/Country
https	//schema.org/Course
https	//schema.org/CourseInstance
https	//schema.org/Courthouse
https	//schema.org/CoverArt
https	//schema.org/CovidTestingFacility
https	//schema.org/CreateAction
https	//schema.org/CreativeWork
https	//schema.org/CreativeWorkSeason
https	//schema.org/CreativeWorkSeries
https	//schema.org/CreditCard
https	//schema.org/Crematorium
https	//schema.org/CriticReview
https	//schema.org/CurrencyConversionService
https	//schema.org/DDxElement
https	//schema.org/DanceEvent
https	//schema.org/DanceGroup
https	//schema.org/DataCatalog
https	//schema.org/DataDownload
https	//schema.org/DataFeed
https	//schema.org/DataFeedItem
https	//schema.org/DataType
https	//schema.org/Dataset
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https	//schema.org/UserInteraction
https	//schema.org/UserLikes
https	//schema.org/UserPageVisits
https	//schema.org/UserPlays
https	//schema.org/UserPlusOnes
https	//schema.org/UserReview
https	//schema.org/UserTweets
https	//schema.org/VacationRental
https	//schema.org/Vehicle
https	//schema.org/Vein
https	//schema.org/Vessel
https	//schema.org/VeterinaryCare
https	//schema.org/VideoGallery
https	//schema.org/VideoGame
https	//schema.org/VideoGameClip
https	//schema.org/VideoGameSeries
https	//schema.org/VideoObject
https	//schema.org/VideoObjectSnapshot
https	//schema.org/ViewAction
https	//schema.org/VirtualLocation
https	//schema.org/VisualArtsEvent
https	//schema.org/VisualArtwork
https	//schema.org/VitalSign
https	//schema.org/Volcano
https	//schema.org/VoteAction
https	//schema.org/WPAdBlock
https	//schema.org/WPFooter
https	//schema.org/WPHeader
https	//schema.org/WPSideBar
https	//schema.org/WantAction
https	//schema.org/WarrantyPromise
https	//schema.org/WarrantyScope
https	//schema.org/WatchAction
https	//schema.org/Waterfall
https	//schema.org/WearAction
https	//schema.org/WearableMeasurementTypeEnumeration
https	//schema.org/WearableSizeGroupEnumeration
https	//schema.org/WearableSizeSystemEnumeration
https	//schema.org/WebAPI
https	//schema.org/WebApplication
https	//schema.org/WebContent

https	//schema.org/WebPage
https	//schema.org/WebPageElement
https	//schema.org/WebSite
https	//schema.org/WholesaleStore
https	//schema.org/WinAction
https	//schema.org/Winery
https	//schema.org/WorkBasedProgram
https	//schema.org/WorkersUnion
https	//schema.org/WriteAction
https	//schema.org/Zoo

5)

When RDFS inferencing is disabled, the triple store exposes only the **explicit triples** contained in the ontology, without generating any additional subclass relationships through RDFS rules. Therefore, the transitivity of `rdfs:subClassOf` is *not* applied at the store level.

To compute all subclasses (direct and indirect) of `schema:CollegeOrUniversity` purely from the explicit data, the transitive closure must be expressed at the **SPARQL query level** using a property path. The expression `rdfs:subClassOf+` instructs SPARQL to follow one or more explicitly stated subclass links, without relying on any inference engine.

The correct SPARQL query is:

```
PREFIX schema: <https://schema.org/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT DISTINCT ?subclass
WHERE {
  ?subclass rdfs:subClassOf+ schema:CollegeOrUniversity .
}
ORDER BY ?subclass
```

This query performs a *syntactic* traversal of explicit `rdfs:subClassOf` triples and therefore retrieves all explicitly reachable subclasses of `schema:CollegeOrUniversity`, even when inferencing is completely disabled.

<i>All queries are shown below:</i>
--

1a)

```
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX dbp: <http://dbpedia.org/property/>
PREFIX dbr: <http://dbpedia.org/resource/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT DISTINCT ?wine ?wineLabel ?region ?regionLabel
WHERE {
  {
    # CASE 1: wineRegion → region in Greece
    ?wine dbo:wineRegion ?region .
    ?region dbo:country dbr:Greece .
  }
  UNION
  {
    # CASE 2: typed as Wine, Greek country, region/ origin multiple properties
    ?wine rdf:type dbo:Wine .
    ?wine dbo:country dbr:Greece .
    ?wine (dbo:region | dbp:region | dbo:origin | dbp:origin) ?region .
    ?region dbo:country dbr:Greece .
  }
  # labels
  OPTIONAL {
    ?wine rdfs:label ?wineLabel .
    FILTER (lang(?wineLabel) = "en")
  }
  OPTIONAL {
    ?region rdfs:label ?regionLabel .
    FILTER (lang(?regionLabel) = "en")
  }
}
```

```
}
```

```
ORDER BY ?wineLabel
```

1b)

```
PREFIX dbo: <http://dbpedia.org/ontology/>
```

```
PREFIX dbp: <http://dbpedia.org/property/>
```

```
PREFIX dbr: <http://dbpedia.org/resource/>
```

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
```

```
SELECT ?pm ?name
```

```
    (GROUP_CONCAT(DISTINCT ?partyLabel; separator=", ") AS ?parties)
```

```
    (GROUP_CONCAT(DISTINCT ?univLabel; separator=", ") AS ?universities)
```

```
WHERE {
```

```
    # Prime ministers of Greece: persons linked from the list page
```

```
    dbr:List_of_prime_ministers_of_Greece dbo:wikiPageWikiLink ?pm .
```

```
    ?pm a dbo:Person .
```

```
    # English name
```

```
    OPTIONAL {
```

```
        ?pm rdfs:label ?name .
```

```
        FILTER (lang(?name) = "en")
```

```
    }
```

Political parties

```
OPTIONAL {  
    ?pm dbp:party ?party .  
    OPTIONAL {  
        ?party rdfs:label ?partyLabel .  
        FILTER (lang(?partyLabel) = "en")  
    }  
}
```

Universities (alma mater)

```
OPTIONAL {  
    ?pm dbp:almaMater ?univ .  
    OPTIONAL {  
        ?univ rdfs:label ?univLabel .  
        FILTER (lang(?univLabel) = "en")  
    }  
}  
}  
  
GROUP BY ?pm ?name  
ORDER BY ?name
```

2a)

```
PREFIX gag: <http://geo.linkedopendata.gr/gag/ontology/>  
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>  
SELECT ?name ?population
```



```

WHERE {
  ?d rdf:type gag:Δήμος ;
  gag:έχει_επίσημο_όνομα ?name ;
  gag:έχει_πληθυσμό ?population .
}
ORDER BY ?name

```

2b)

```

PREFIX gag: <http://geo.linkedopendata.gr/gag/ontology/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

```

```

SELECT ?regionName ?unitName ?municipalityName
WHERE {
  ?region rdf:type gag:Περιφέρεια ;
    gag:έχει_επίσημο_όνομα ?regionName .

  ?unit rdf:type gag:Περιφερειακή_Ενότητα ;
    gag:ανήκει_σε ?region ;
    gag:έχει_επίσημο_όνομα ?unitName .

  ?municipality rdf:type gag:Δήμος ;
    gag:ανήκει_σε ?unit ;
    gag:έχει_επίσημο_όνομα ?municipalityName .
}
ORDER BY ?regionName ?unitName ?municipalityName

```

2c)

```
PREFIX gag: <http://geo.linkedopendata.gr/gag/ontology/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

SELECT ?municipalityName ?population ?unitName
WHERE {
  # Regions – pick only the Region of Peloponnese
  ?region rdf:type gag:Περιφέρεια ;
    gag:έχει_επίσημο_όνομα ?regionName .

  FILTER( CONTAINS( STR(?regionName), "ΠΕΛΟΠΟΝΝΗΣΟΥ" ) )

  # Regional units of Peloponnese
  ?unit rdf:type gag:Περιφερειακή_Ενότητα ;
    gag:ανήκει_σε ?region ;
```

gag:έχει_επίσημο_όνομα ?unitName .

Municipalities belonging to each regional unit

?municipality rdf:type gag:Δήμος ;

gag:ανήκει_σε ?unit ;

gag:έχει_επίσημο_όνομα ?municipalityName ;

gag:έχει_πληθυσμό ?population .

Population filter > 5000

FILTER(xsd:integer(?population) > 5000)

}

ORDER BY ?municipalityName ?unitName

2d)

PREFIX gag: <http://geo.linkedopendata.gr/gag/ontology/>

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT ?municipalityName

WHERE {

?region rdf:type gag:Περιφέρεια ;

gag:έχει_επίσημο_όνομα ?regionName .

FILTER(CONTAINS(STR(?regionName), "ΠΕΛΟΠΟΝΝΗΣΟΥ"))

?unit rdf:type gag:Περιφερειακή_Ενότητα ;

gag:ανήκει_σε ?region .

?municipality rdf:type gag:Δήμος ;

gag:ανήκει_σε ?unit ;

gag:έχει_επίσημο_όνομα ?municipalityName .

FILTER NOT EXISTS { ?municipality gag:έχει_έδρα ?seat }

}

ORDER BY ?municipalityName

2e)

PREFIX gag: <http://geo.linkedopendata.gr/gag/ontology/>

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT DISTINCT ?municipalityName ?admName

WHERE {

```

?pelRegion rdf:type gag:Περιφέρεια ;
    gag:έχει_επίσημο_όνομα ?regionName .
FILTER(CONTAINS(STR(?regionName), "ΠΕΛΟΠΟΝΝΗΣΟΥ"))

?municipality rdf:type gag:Δήμος ;
    gag:έχει_επίσημο_όνομα ?municipalityName .
?municipality gag:ανήκει_σε+ ?pelRegion .

?municipality gag:ανήκει_σε+ ?admUnit .
?admUnit gag:έχει_επίσημο_όνομα ?admName .
}

ORDER BY ?municipalityName ?admName

```

PREFIX gag: <http://geo.linkedopendata.gr/gag/ontology/>

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

SELECT ?regionName ?regionMunicipalities ?unitName ?unitMunicipalities

WHERE {

Region name

?region rdf:type gag:Περιφέρεια ;

gag:έχει_επίσημο_όνομα ?regionName .

Subquery: number of municipalities per region

{

SELECT ?region (COUNT(DISTINCT ?mun) AS ?regionMunicipalities)

WHERE {

?unit rdf:type gag:Περιφερειακή_Ενότητα ;

gag:ανήκει_σε ?region .

?mun rdf:type gag:Δήμος ;

gag:ανήκει_σε ?unit .

}

GROUP BY ?region

}

Subquery: number of municipalities per regional unit (within each region)

{

SELECT ?region ?unitName (COUNT(DISTINCT ?mun2) AS ?unitMunicipalities)

WHERE {

?unit rdf:type gag:Περιφερειακή_Ενότητα ;

gag:ανήκει_σε ?region ;

gag:έχει_επίσημο_όνομα ?unitName .

?mun2 rdf:type gag:Δήμος ;

gag:ανήκει_σε ?unit .

```

    }
    GROUP BY ?region ?unitName
  }
}
ORDER BY ?regionName ?unitName

```

2g)

```

PREFIX gag: <http://geo.linkedopendata.gr/gag/ontology/>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>

SELECT ?parent ?parentName ?parentPop
      (SUM(xsd:integer(?childPop)) AS ?childrenPop)
WHERE {
  # Parent administrative unit with stated population (level L+1)

```

```

?parent gag:έχει_πληθυσμό ?parentPop .

OPTIONAL { ?parent gag:έχει_επίσημο_όνομα ?parentName . }

# Child administrative units of level L that belong to this parent
?child gag:ανήκει_σε ?parent ;

    gag:έχει_πληθυσμό ?childPop .
}

GROUP BY ?parent ?parentName ?parentPop
HAVING ( xsd:integer(?parentPop) != SUM(xsd:integer(?childPop)) )

ORDER BY ?parentName

```

2h)

```

PREFIX gag: <http://geo.linkedopendata.gr/gag/ontology/>

PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

```



```
SELECT DISTINCT ?daName
WHERE {
    ?da rdf:type gag:Αποκεντρωμένη_Διοίκηση ;
        gag:έχει_επίσημο_όνομα ?daName .

    ?unit1 rdf:type gag:Περιφερειακή_Ενότητα ;
        gag:ανήκει_σε ?da .
    ?unit2 rdf:type gag:Περιφερειακή_Ενότητα ;
        gag:ανήκει_σε ?da .
    ?unit3 rdf:type gag:Περιφερειακή_Ενότητα ;
        gag:ανήκει_σε ?da .

    FILTER(?unit1 != ?unit2)
    FILTER(?unit1 != ?unit3)
    FILTER(?unit2 != ?unit3)
}
ORDER BY ?daName
```

3a)

PREFIX schema: <https://schema.org/>

PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT DISTINCT ?subclass

WHERE {

 ?subclass rdfs:subClassOf+ schema:CollegeOrUniversity .

}

3b)

PREFIX schema: <https://schema.org/>

PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT DISTINCT ?superclass

WHERE {

 schema:CollegeOrUniversity rdfs:subClassOf+ ?superclass .

}

3c)

PREFIX schema: <https://schema.org/>

PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT DISTINCT ?property ?domain

WHERE {

CollegeOrUniversity and all its superclasses

schema:CollegeOrUniversity rdfs:subClassOf* ?domain .

Properties whose (schema.org) domain includes one of these classes

?property schema:domainIncludes ?domain .

}

ORDER BY ?domain ?property

3d)

PREFIX schema: <https://schema.org/>

PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT DISTINCT ?class

WHERE {

{

subclasses at distance 1 from Thing

?class rdfs:subClassOf schema:Thing .

}

UNION

{

subclasses at distance 2 from Thing

?class rdfs:subClassOf ?mid .

?mid rdfs:subClassOf schema:Thing .

}

```
    FILTER (?class != schema:Thing)
}
ORDER BY ?class
```

3e)

```
PREFIX schema: <https://schema.org/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT DISTINCT ?subclass
WHERE {
    ?subclass rdfs:subClassOf+ schema:CollegeOrUniversity .
}
```

ORDER BY ?subclass

